

Dynamic Analysis and Stability Enhancement of a Weak Grid-Tied Hybrid System Integrating PV, Full-Scale Wind Turbine, and Battery Storage

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ABSTRACT Hybrid generation systems integrating photovoltaics (PVs), full-scale wind turbines (WTs), and battery energy storage systems (BESSs) have garnered substantial interest for utility-scale grid-tied applications due to their enhanced efficiency and reliability. However, in a hybrid system integrating these resources at the dc-side of a voltage-source converter, the dc-link stability remains largely unexplored in existing literature, particularly under weak grid conditions and across different operational regions of the PV, WT, and BESS. This article presents a comprehensive small-signal stability analysis of a hybrid PV-WT-BESS system using detailed state-space and incremental impedance models. The analysis reveals that the hybrid system exhibits low-frequency oscillation instability in weak grid conditions when the PV operates in the current-limited region. A novel active damping strategy is introduced to mitigate these instabilities, effectively suppressing negative interaction dynamics and improving overall system stability; this is achieved by repositioning unstable eigenmodes and reshaping the dc-link dynamics to comply with the Nyquist stability criterion. The proposed compensation technique offers several key advantages: 1) it is simple yet effective and can be designed using linear analysis methods; 2) it ensures stable operation under varying grid conditions without altering steady-state performance; 3) it enhances the low-voltage ride-through capability; and 4) it eliminates the need for additional measurement sensors, thereby simplifying the implementation and reducing costs compared to alternative stabilization methods. Offline and real-time software-in-the-loop simulations validate the accuracy of the developed models and dynamic interactions, as well as the effectiveness of the proposed active stabilization approach across typical operating conditions.

INDEX TERMS Active damping control, battery energy storage, dc-link stability, hybrid renewable generation, impedance-based stability analysis, photovoltaic (PV) systems, small-signal analysis, voltage-source converters, weak grid, wind turbines (WTs).

NOMENCLATURE

A. Grid-Tied VSI Variables and Parameters

ω, e_g, i_g
 P_{inv}

Grid angular frequency, voltage, and current.
Total output power contribution from external PV, WT, and BESS units delivered to the lossless VSI converter.

R_g, L_g	Grid resistance and inductance.
R_f, L_f	AC filter resistance and inductance.
r_{on}	VSI's IGBTs on-state resistance.
i, v_i	VSI ac terminal output current and voltage.
j	Imaginary number unit.
N	Turns ratio of the step-up transformer.
R_{trp}, L_{trp}	Step-up transformer's primary resistance and inductance.
R_{trs}, L_{trs}	Step-up transformer's secondary resistance and inductance.
v, C_f	AC filter voltage and shunt capacitor.
P_{pcc}	Delivered ac power at the point of common coupling.
P_{dc}	DC power delivered to the VSI.
G^δ, G^v, G^i	PI-controllers for the SRF-PLL, PCC voltage, and VSI terminal ac current control loops.
g_p^x, g_i^x	Proportional and integral gains of G^x , where $x = \delta, v, i$.

B. PV Parameters and Variables

V_{pv}, I_{pv}	PV generator's terminal voltage and current.
I_{sc}, I_{ph}	PV module-based PSDM model's short-circuit and photo-generated currents.
I_{rs}, I_D	PV module-based PSDM model's reverse saturation current and diode current.
V_T, V_D	PV module-based PSDM model's junction thermal and diode voltages.
T, I_{irr}	PV cell temperature and solar irradiance levels.
k_B	Boltzmann's constant.
A	Ideality factor of the diode.
q	Elementary electric charge unit.
α_i	Temperature coefficient of the short-circuit current.
R_s, R_{sh}	PV module-based PSDM model's series and shunt resistances.
R_{dc}, L_{dc}	Resistance and inductance of the dc transmission cable.
V_{dc}, C_{dc}	DC-link voltage and equivalent capacitor.
C_{pv}	PV-side dc-link capacitor.
P_{pv}^{ext}	External power generation delivered by the PV system.

C. WT Variables and Parameters

v_{WT}, ρ	Wind speed and air density.
β, R	Blade's pitch angle and radius.
λ	Tip-speed ratio.
H_{WT}	Wind turbine's inertia coefficient.
T_{WT}, ω_{WT}	Wind turbine's torque and angular speed.
$K_{sh}, \theta_{sh}, T_{sh}$	Shaft stiffness, angle, and torque.
ω_r^{mech}	Rotor's rotational mechanical speed.
D_{sh}, ω_b	Shaft's mutual damping and base speed.
P	Number of pole pairs.
ω_s^{magn}	Stator's rotational magnetic field speed.
L_s, R_s	Stator-winding's inductance and resistance.
i_s, v_s	Stator current and voltage.

B_r, J_r	Viscous friction coefficient and rotor inertia constant of the PMSG.
T_m, T_e	Mechanical and electromagnetic torques of the PMSG.
λ_{pm}	Rotor magnetic flux linkage.
P_m	Mechanical input power to the PMSG.
P_s	Electrical output power of the stator.
C_{WT}	Wind turbine dc-link capacitor.
P_{WT}^{ext}	External power generation delivered by the wind turbine system.
K^i, K^s	PI-controllers for the PMSG's stator current and rotor speed control loops.
k_p^x, k_i^x	Proportional and integral gains of K^x where $x = i, s$.

D. BESS Parameters and Variables

V_{bat}, I_{bat}	Battery voltage and current.
Z_{sd}, E_{sd}	Self-discharge impedance and voltage.
$I_{R_1^{bat}}$	Current passing through battery resistance R_1^{bat} .
C_{bat}	BESS-side dc-link capacitor.
P_{bat}^{ext}	External charging/discharging power of the battery.
L^v, L^i	PI-controllers for dc-link voltage and battery current control loops.
l_p^x, l_i^x	Proportional and integral gains of L^x where $x = v, i$.

E. Subscript

d, q	Direct and quadrature axes.
dq	Direct-Quadrature reference frame.

F. Superscript

g	System's measured signals in the dq -grid reference frame.
c	Controllers' output signals in the dq -converter reference frame.
$*$	Nominal/Reference value.
$\circ$	Steady-state operating point.

I. INTRODUCTION

Hybrid photovoltaic (PV), wind turbine (WT), and battery energy storage systems (BESSs) offer a promising solution for enhancing the integration of renewable energy systems by combining the complementary characteristics of solar and wind power generation, along with energy storage capabilities. This combination improves generation efficiency and reliability [1]. As of early 2024, global installed capacity shares for PV and WT generation have reached 37% and 26%, respectively, out of a total renewable energy capacity of 3870 GW worldwide [2]; this marks a substantial increase from their respective shares of 15% and 23% out of a total capacity of 2006 GW reported in 2016 [3]. In addition, BESS integration has experienced exponential growth, expanding from 9 GW/17 GWh in 2018 to a projected 1095 GW/2850

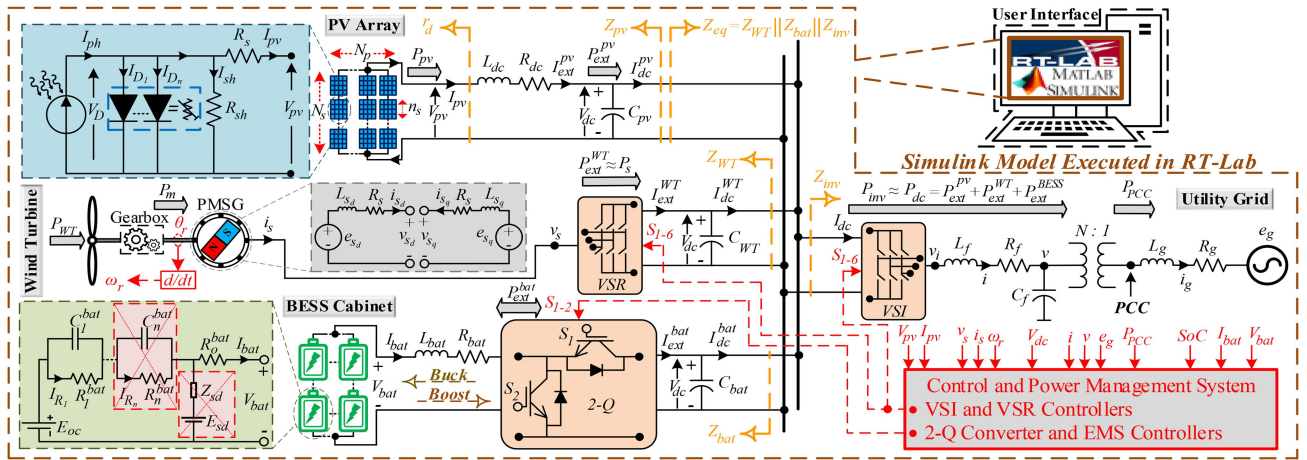


FIGURE 1. Configuration of the grid-tied hybrid PV-WT-BESS system and its implementation on the OPAL-RT simulation platform.

GWh by 2040 [4], highlighting the rapid adoption of battery technology for energy storage systems (ESSs).

The hybrid system depicted in Fig. 1 is considered in this work. This configuration enhances the energy conversion efficiency. The system incorporates PV, WT, and BESS units, all connected at the dc-link and interfaced with the grid through a two-level voltage-source inverter (VSI). Grid-tied PV systems are classified into single- and double-stage configurations [5]. The former integrates a peak-power-point-tracking (PPPT) algorithm into the interfacing grid-tied VSI to maximize power extraction from the PV generator at the optimal peak-power-point (PPP), and the latter utilizes an intermediate dc-dc converter for this purpose [6], [7]. It has been reported that single-stage PV systems provide an improved design in terms of lower cost, smaller size, and reduced losses, particularly for large-scale PV systems; therefore, the single-stage PV system is adopted in this work. For utility-scale grid-tied WT systems, the most common types employ doubly-fed induction generators (WT-DFIGs) and permanent magnet synchronous generators (WT-PMSGs) [8], [9]. The WT-PMSG exhibits higher efficiency, greater flexibility in power quality regulation, and superior grid fault ride-through capabilities [9] than the WT-DFIG; therefore, a WT-PMSG is adopted in the hybrid system under study. Similarly, several types of ESS for grid-tied applications are discussed in the literature [10], [11], with lithium-ion BESS being the most widely adopted for utility-scale grid-tied systems due to its reliability, compact size, lower cost, high efficiency, and reduced maintenance requirements [12]. Consequently, a lithium-ion BESS is considered in this work.

As shown in Fig. 1, the dc-link is interfaced with the grid via a VSI, and the grid stiffness significantly affects the synchronization and dynamic stability of the interfacing VSI [13]. Grid stiffness is typically defined by the short-circuit-ratio (SCR), which represents the ratio of the rated three-phase short-circuit power at the point-of-common-coupling (PCC) to the rated dc-power of the interconnected VSI [14]. The grid is considered strong when $SCR > 3$, weak when $2 \leq SCR \leq$

3, and very weak when $SCR < 2$, with the most challenging scenario occurring in extremely weak grids where $SCR = 1$ [15]. This work focuses on weak grid conditions, specifically in the $1 < SCR < 3$ range. Based on grid synchronization approaches, VSIs are generally categorized into grid-following and grid-forming inverters [16], [17]. Grid-following VSIs primarily rely on measuring PCC voltage for synchronization with the grid, using a synchronous-reference-frame phase-locked-loop (SRF-PLL) to determine the phase angle and frequency of the PCC voltage, which is then utilized by the VSI controllers [9], [15]. On the other hand, grid-forming VSIs mainly employ active power-frequency control for grid synchronization, where the PCC voltage is regulated by a voltage controller [18], [19], [20].

Recent research has introduced different PLL-less power-synchronized grid-following and grid-forming VSIs, primarily based on direct power control, to overcome synchronization challenges and improve stability in weak grid systems [16]. However, the instability problems identified in this study are not dominantly related to the PLL. Instead, these problems are primarily driven by the interaction between weak grid impedance and the different operating regions of the PV, WT, and BESS units, causing dc-link oscillations. Therefore, the conventional SRF-PLL is considered for synchronization in the proposed hybrid PV-WT-BESS grid-following system.

A. STATE-OF-THE-ART

Integrating grid-tied PV, WT, and BESS systems requires careful consideration of dynamic stability and efficient energy management. As summarized in Table 1, existing literature explores the energy management problem through two key perspectives: 1) optimization and energy management strategies—such as forecasting, economic dispatch, energy storage control, demand response, and multiobjective optimization—which play a crucial role in maximizing system efficiency and economic viability [11], [21], [22]; and 2) dynamic interaction analysis, involving mathematical modeling, small-signal stability analysis, grid interaction, and

TABLE 1. Comparison With the State-of-the-Art

Ref. No.	Grid-Tied Hybrid System					Interaction Dynamic Stability Analysis				
	PV		WT		BESS	Stability Assessment	Stability Enhancement	Modeling Analysis		
	PPP	OFF-PPP	PPP	OFF-PPP	Ch, SB, DCh			LSSM	ISAM	TDNLM
[1]	✓	×	✓	×	×	✓	✓	✓	×	✓
[4]	✓	✓	✓	✓	✓	×	×	✓	×	✓
[5]–[6]	✓	✓	×	×	×	✓	×	×	✓	✓
[7], [13]–[15], [34]	✓	✓	×	×	×	✓	✓	✓	×	✓
[9]	✓	✓	✓	✓	×	✓	✓	✓	×	✓
[8], [17]	×	×	✓	×	×	✓	✓	×	✓	✓
[18]	✓	×	×	×	×	✓	✓	✓	×	✓
[19]	×	×	✓	×	✓	✓	✓	✓	×	✓
[24]	✓	×	✓	×	✓	×	×	×	×	✓
[21]–[23], [25]–[28]	✓	×	✓	×	✓	×	×	×	×	✓
[29]–[30]	✓	×	✓	×	✓	×	×	×	✓	✓
[20], [31]–[33]	✓	×	✓	×	✓	✓	✓	×	×	✓
This Article	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

control design for PV, WT, and BESS converters, which are critical for ensuring grid stability under varying operating conditions [1], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34]. Combining these energy management strategies with dynamic stability analysis enables grid-tied hybrid PV-WT-BESS systems to achieve both stability and efficiency, thereby significantly contributing to the reliability and sustainability of the power grid [1].

A supervisory power control strategy for a grid-tied hybrid generation system integrating two-stage PV, WT, and BESS units is introduced in [23], ensuring reliable and efficient power transfer in grid-tied applications. The study in [24] presents a small-signal stability analysis of hybrid single-stage PV and WT generators connected to an isolated load and various ESS units, including BESS. A coordination control algorithm proposed in [25] facilitates smooth power transfer between ac and dc microgrid systems of multigeneration and storage units, ensuring stability under various generation and load conditions. In [1], the stability of large-scale hybrid power systems, which integrate a WT-PMSG, two-stage PV farm, and supercapacitor energy storage, is enhanced. By eliminating the PV converter and simplifying control loops in [26], a cost-efficient hybrid grid-tied system that integrates WT-DFIG, single-stage PV, and BESS units is proposed.

Power quality and stability challenges arising from uncertain fluctuations in renewable energy and loading conditions are addressed in [27], [28], [29], [30], [31], [32], and [33]. In [27], a hybrid system with WT-PMSG, two-stage PV, and BESS units is proposed to address power quality issues, such as harmonic distortion and load balancing. In [28], voltage quality challenges caused by fluctuating energy sources and varying power demand are explored in a hybrid system consisting of WT, two-stage PV, and BESS units. A new configuration for a grid-interactive dc microgrid integrating two-stage PV, WT, and hybrid energy storage units is introduced in [29] to effectively manage power fluctuations in PV and WT generation and sudden load disturbances. In [30], a

detailed analysis, modeling, and design of a new configuration for a standalone dc microgrid with multiple two-stage PV, WT, and hybrid energy storage units, including BESS, is proposed, aiming to mitigate high-frequency fluctuations, extend battery lifetime, and reduce the size of the storage units. A grid-tied hybrid system with a single-stage PV, WT-DFIG, and energy storage with enhanced control schemes is proposed in [31] to improve power quality and reduce transient response times. Similarly, a hybrid system consisting of a WT-DFIG, battery, and two-stage PV system is explored in [32], focusing on advanced control techniques to enhance power quality and ensure efficient operation. In [33], a generalized fourth-order filter and proportional resonant control approach for a hybrid system with two-stage PV, WT-PMSG, and BESS units are developed to mitigate power quality issues and ensure stable transitions between grid-tied and off-grid operations.

Instability problems in grid-tied single-stage PV and WT-PMSG systems are primarily examined in off-PPP operational regions in [5], [6], [7], [9], [13], [15], and [34]. The influence of a PV generator's dynamic resistance on stiff and weak grids is investigated in grid-tied PV-only systems, considering reduced dc-link capacitance across all PV operating regions [5], [6]. An active damping method is proposed in [7] to mitigate the interaction dynamics identified in [5] and [6]. Moreover, the impact of the dynamic resistance of different PV generators' equivalent circuits in stiff and weak grid conditions across both PPP and off-PPP regions is examined in [13] for grid-tied PV-only systems. Recent studies [15], [34] indicate that weak grid-tied PV-only systems may experience low-frequency oscillation (LFO), medium-frequency oscillation, and high-frequency oscillation instabilities in off-PPP regions. In [9], an active damping control method is introduced for a hybrid system combining single-stage PV and WT-PMSG units to address interaction dynamics between the high impedance of weak grids and the equivalent dynamic resistance of PV and WT generators in PPP and off-PPP operations. However, this study does not account for the operating

regions of the BESS unit or various system operating scenarios. In contrast, all operational scenarios of a hybrid PV, WT, and BESS system are considered in [4] but primarily focus on developing an aggregated equivalent model for mathematical representation. Nevertheless, it does not assess the system stability or develop stabilization strategies for potential instabilities under specific operating conditions.

B. PROBLEM FORMULATION

Recent comprehensive reviews comparing different PPPT techniques for PV and WT systems have been presented in [35] and [36], respectively, highlighting their advantages and limitations. Due to unpredictable and fluctuating atmospheric conditions, PPPT algorithms for PV and WT systems may encounter challenges in accurately tracking the optimal PPP, which is primarily attributed to their slow response, leading to PV and WT systems operating outside the desired PPP range [9]. Moreover, PV and WT generators may be intentionally adjusted to operate in off-PPP regions through strategic curtailment methods to maintain energy management in hybrid systems [4]. In addition, the response of the energy management systems (EMSs) in grid-tied hybrid sources is typically slow, leading to short-term BESS units operating under various conditions, including charging (Ch), discharging (DCh), and standby (SB) modes [37]. As a result, the dynamic resistance exhibited by PV, WT, and BESS units significantly impacts the overall system's dynamic stability, primarily when operating across different regions of their characteristics and under weak grid conditions. This research gap has not been thoroughly explored in the literature and is, therefore, the central focus of this study.

Existing research on hybrid PV, WT, and BESS systems has extensively explored cost optimization, system sizing, and dynamic stability analysis, primarily under optimal PPP conditions. However, to the best of the authors' knowledge, there remains a gap in the literature regarding a comprehensive assessment of how the dynamic resistances of energy sources influence the stability of utility-scale weak grid-tied hybrid systems. In particular, no prior studies have thoroughly examined the stability implications of integrating single-stage PV, WT-PMSG, and BESS units across their different operational regions, including PPP and off-PPP conditions of the PV and WT and the various operating modes of the BESS. Existing works do not propose stability enhancement strategies tailored to these scenarios. The present study investigates the dynamic stability of a grid-tied hybrid system comprising single-stage PV, WT-PMSG, and BESS units, focusing on stability margins, uncertainty analysis, and active damping strategies. The research further examines the interaction dynamics between the single-stage PV, WT-PMSG, and BESS units on the dc side and the utility grid on the ac side, considering all operational conditions of the energy sources under weak grid connection. Given the inherent limitations of BESS, including low energy density and self-discharge effects, along with the typically slow response of the EMS in grid-tied hybrid systems [37], this study prioritizes the

fast dynamics associated with short-term BESS operation and power management controllers, analyzing their performance under varying EMS-generated setpoints. Moreover, a novel damping control strategy with multiple degrees of freedom is introduced to ensure stable operation across various operating conditions.

C. CONTRIBUTIONS

The contributions of this article to the research field are outlined as follows.

- 1) A comprehensive set of models, including a large-signal time-domain nonlinear model (TDNLM), a small-signal linearized state-space model (LSSM), and an incremental impedance-based stability analysis model (ISAM), is developed for a grid-tied hybrid system integrating a single-stage PV, a WT-PMSG, and a BESS unit. These models are used for an in-depth assessment of the system's dynamics and stability across various operating conditions, including weak grid connection, operation within and beyond the PPP regions of the PV and WT characteristics, and all BESS operating modes.
- 2) The interaction dynamics between the ac and dc sides of the hybrid system's VSI are extensively analyzed, examining the influence of the dynamic resistance of energy sources across all operational zones.
- 3) A novel and efficient active damping strategy is introduced to enhance the stability of the dc-link dynamics under diverse operational conditions, particularly in weak grid connections and across different PV, WT, and BESS operating regions.

The rest of this article is organized as follows. The modeling, control, and power management details of the grid-tied hybrid PV-WT-BESS system are given in Section II. Section III investigates the impact of energy sources' dynamic resistance on system stability, considering different PV, WT, and BESS operating regions. Section IV introduces an active damping approach to stabilize dc-link dynamics, supported by an impedance-based stability analysis. Section V evaluates the effectiveness of the proposed stabilization method through software-in-the-loop (SIL) real-time simulations. Finally, Section VI concludes this article.

II. MODELING, CONTROL, AND POWER MANAGEMENT OF THE GRID-TIED HYBRID PV-WT-BESS SYSTEM

The configuration of the grid-tied hybrid system consisting of a single-stage PV, a WT-PMSG, and a BESS is shown in Fig. 1. In this setup, the dc-link bus of the VSI interfaces the WT generator and the battery bank through a voltage-source rectifier (VSR) and a two-quadrant (2-Q) bi-directional dc-dc converter, respectively. In contrast, the PV generator is directly connected to the dc-link bus via a dc cable [38], eliminating the need for a dc-dc converter [9]. This setup aims to reduce losses and costs, particularly in utility-scale applications [13], [15]. On the ac-side of the VSI, a filter is utilized to mitigate the high-frequency switching harmonics

[39], and a step-up transformer is integrated at the PCC to increase voltage level and minimize power losses [7].

The VSR is utilized to convert the WT's generated power into the dc-bus [9], [38], and the bidirectional 2-Q converter is employed to manage power flow between the BESS and the dc-bus in both forward and reverse directions [4]. The VSI is utilized to convert the dc-link power into the grid. The control loops for the VSI, VSR, and 2-Q converters are divided into two main stages: inner and outer loops [39]. The inner loops control the VSI current, the stator current of the PMSG, and the battery current. Meanwhile, the outer loops control the PCC voltage, the rotor speed of the PMSG, and the dc-link voltage. To ensure a rapid response and stable operation, the inner loops' bandwidths are set to approximately 15% of the switching frequency of the VSI, VSR, and 2-Q converters, and the outer loops are typically designed with bandwidths of around 15% of the inner loops' bandwidths [13], [39]. All control loops maintain a minimum phase margin of 45° to ensure stable operation and prevent potential instabilities [15].

A detailed overview of the system's modeling, control, and power management strategies is provided in the following sections.

A. MODELING AND CONTROL OF THE GRID-TIED VSI

The grid impedance, denoted as $Z_g = R_g + j\omega^\circ L_g$, is defined based on the SCR level and the reactance-to-resistance (X/R) ratio, as provided in (1a)–(1b) [15]

$$L_g = \left[e_g^2 / \left(P_{\text{pcc}}^* \text{SCR} \right) \right] / \left[\omega \sqrt{1 + [1/(X/R)]^2} \right] \quad (1a)$$

$$R_g = \left[e_g^2 / \left(P_{\text{pcc}}^* \text{SCR} \right) \right] / \sqrt{1 + (X/R)^2}. \quad (1b)$$

In the dq -grid reference frame, the mathematical representations of the combined grid and transformer impedances, along with the LC filter, are expressed by (2)–(4)

$$L_f \frac{d}{dt} (i_{dq}^g) + j\omega L_f i_{dq}^g + (R_f + r_{on}) i_{dq}^g = v_{dq}^g - v_{dq}^g \quad (2)$$

$$L_T \frac{d}{dt} (i_{gdq}^g) + j\omega L_T i_{gdq}^g + R_T i_{gdq}^g = \left(\frac{1}{N} \right) v_{dq}^g - e_{gdq}^g \quad (3)$$

$$C_f \frac{d}{dt} (v_{dq}^g) + j\omega C_f v_{dq}^g = i_{dq}^g - \left(\frac{1}{N} \right) i_{gdq}^g \quad (4)$$

where d/dt is the state derivative with respect to time; $R_{tr} = R_{tr_s} + R_{tr_p}/N^2$ and $L_{tr} = L_{tr_s} + L_{tr_p}/N^2$ represent the expressions of the equivalent transformer resistance and inductance referred to the secondary side, respectively; L_T is the total inductance comprising both L_g and L_{tr} ; and R_T is the total resistance, combining R_g and R_{tr} .

The SRF-PLL synchronizes the VSI with the utility grid [39]. To achieve this, the SRF-PLL controller $G^\delta = g_p^\delta + g_i^\delta/s$ is used, as described in (5a)–(5c) and shown in Fig. 2, to set the q -component of the ac capacitor voltage (v_q^c) to zero and generate the synchronization angle (δ^c), which is required for

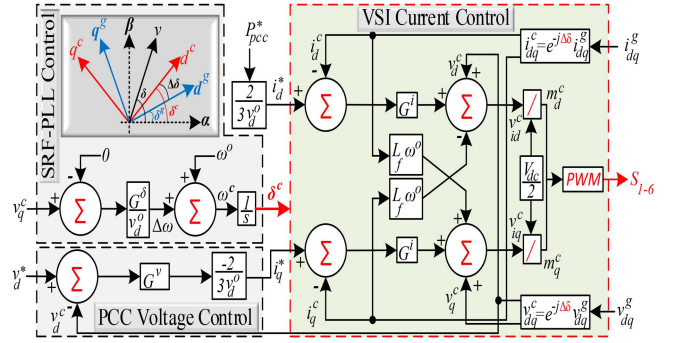


FIGURE 2. Schematic diagram of the VSI vector control for the grid-side circuit.

dq frame transformations [7]

$$\omega^c = \Delta\omega + \omega^\circ \quad (5a)$$

$$\omega^c = \frac{d}{dt} (\delta^c) \quad (5b)$$

$$\Delta\omega = \left(g_p^\delta + \frac{g_i^\delta}{s} \right) \left(\frac{v_q^c}{v_d^c} \right). \quad (5c)$$

The conversion of the system's measured signals from the dq -grid frame, expressed as $f_{dq}^g = e^{-j\delta^g} f_{\alpha\beta}^g$, to the controllers' output signals in the dq -converter frame, expressed as $f_{dq}^c = e^{-j\delta^c} f_{\alpha\beta}^g$, and vice versa, is a crucial and indispensable step in accurately capturing the influence of the SRF-PLL on system dynamics, especially under weak grid conditions [15], [40]. This transformation between frames is illustrated in Fig. 2 and mathematically described in the following equation [9]:

$$f_{dq}^c = e^{-j\delta^c} \left(e^{j\delta^g} f_{dq}^g \right) = e^{-j(\delta^c - \delta^g)} f_{dq}^g = e^{-j\Delta\delta} f_{dq}^g. \quad (6)$$

The d -component of the reference current, expressed as $i_d^* = P_{\text{pcc}}^* / (1.5v_d^o)$, determines the desired power that needs to be injected into the grid. On the other hand, the q -component of the reference current, denoted as i_q^* , is generated by regulating the d -component of the ac voltage (v_d) to the desired value (v_d^*) through the PCC voltage controller $G^v = g_p^v + g_i^v/s$, as follows:

$$i_q^* = \frac{-1}{1.5v_d^o} \left(g_p^v + \frac{g_i^v}{s} \right) (v_d^* - v_d) \quad (7)$$

where $v_d^o = \sqrt{2/3}v_{\text{rms}}$ represents the peak value of the PCC voltage.

The ac current controller, $G^i = g_p^i + g_i^i/s$, is utilized to regulate the dq -component of the ac current (i_{dq}^c) to its desired value (i_{dq}^*), as shown in (8) and Fig. 2. This controller generates the modulation signals, $m_{dq} = v_{dq}^c / (V_{dc}/2)$. A conventional pulse-width modulation (PWM) scheme uses the generated modulation signals (m_{dq}) to produce the switching function for the VSI's switches [39]

$$v_{dq}^c = v_{dq}^c + j\omega^\circ L_f i_{dq}^c + \left(g_p^i + \frac{g_i^i}{s} \right) (i_{dq}^* - i_{dq}^c). \quad (8)$$

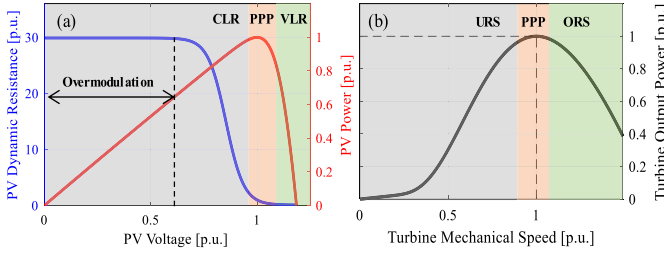


FIGURE 3. Characteristics of the PV and WT Generators. (a) PV generator - $P_{pv} - V_{pv}$ and $r_d - V_{pv}$ at $I_{irr} = 1.0 \text{ kW/m}^2$. (b) WT generator - $P_{wt} - \omega_{wt}$ at $v_{wt} = 12 \text{ m/s}$.

In (8), v_{dq}^c and $j\omega^\circ L_f i_{dq}^c$ represent the feed-forward and decoupling terms, respectively.

B. PV GENERATOR MODEL

Fig. 1 illustrates the configuration of the PV generator, which involves the interconnection of multiple parallel strings (N_p). Each string consists of N_s modules connected in series, and within each module, n_s PV cells are connected in series [7]. For this research study, the practical single-diode model (PSDM), a widely used equivalent circuit, is preferred due to its simplicity and reasonably accurate performance compared to other more complex PV equivalent models introduced in the literature [9], [39]. The power delivered by the PV generator (P_{pv}) is defined as shown in (9a)–(9e) [25], [38]

$$P_{pv} = V_{pv} I_{pv} \quad (9a)$$

$$I_{pv} = N_p I_{ph} - N_p I_D - N_p (V_D / R_{sh}) \quad (9b)$$

$$V_D = [V_{pv} + (N_s / N_p) R_s I_{pv}] / N_s \quad (9c)$$

$$I_{ph} = (I_{irr} / I_{irr}^*) I_{sc} [1 + \alpha_i (T - T^*)] \quad (9d)$$

$$I_D = I_{rs} [\exp \{(qV_D) / (n_s A k_B T)\} - 1]. \quad (9e)$$

The static resistance (r_s) of the PV generator is determined by the ratio of V_{pv} to I_{pv} [6], [34]. To accurately assess the dynamic performance of the grid-tied hybrid system, it is essential to consider the dynamic resistance (r_d) of the PV generator in both the PPP and off-PPP regions, i.e., the voltage-limited-region (VLR) and the current-limited-region (CLR) [6], [7]. This consideration is necessary due to the nonlinear behavior of the p - n junction diode [7], as depicted in Fig. 3(a), and the rapid fluctuations in operating points over time [35]. The dynamic resistance for any PV equivalent circuit model can be calculated as the negative reciprocal of the derivative of I_{pv} with respect to V_{pv} , and it is expressed by (10) for the PV generator-based PSDM [13]

$$r_d = \frac{1 + \frac{R_s}{R_{sh}} + \frac{I_{rs} R_s}{n_s A V_T} \exp \left\{ \frac{V_D}{n_s A V_T} \right\}}{\left(\frac{1}{(N_s / N_p) R_{sh}} + \frac{N_p I_{rs}}{N_s n_s A V_T} \exp \left\{ \frac{V_D}{n_s A V_T} \right\} \right)}. \quad (10)$$

The dynamic equation of the dc cable connecting the PV generator to the VSI's dc-side is given by [38]

$$L_{dc} \frac{d}{dt} (I_{pv}) + R_{dc} I_{pv} = V_{pv} - V_{dc} \quad (11)$$

where V_{pv} is the PV voltage given by $V_{pv} = P_{pv} / I_{pv}$, and it can also be expressed as $V_{pv} = -r_d I_{pv}$ to study the impact of the PV's dynamic resistance [9].

It is necessary to select V_{dc} to be greater than twice the peak value of v_i to prevent the VSI from operating in the nonlinear mode of operation, i.e., the overmodulation region, and to avoid generating undesirable low-order harmonic components in the VSI output voltage spectrum [9], [15].

In PV systems, reverse power flow can occur when the PV panels are not generating electricity, such as during the night or when shaded, potentially damaging the PV panels if not appropriately managed [41]. The following measures are commonly implemented to protect the PV system against reverse currents: 1) blocking diodes, also referred to as bypass diodes or anti-reverse diodes, are typically integrated into the design of PV modules by manufacturers, i.e., placed in parallel with PV cells, to allow current flow in one direction from the PV plant to the dc-link, while preventing reverse current flow through the PV generator [42], [43]; 2) dc-link voltage controller is commonly designed to manage the error between the reference and actual values of the PV generator; this ensures that the generated current consistently feeds into the dc-link in a positive direction, regardless of operating conditions—at night or in daylight [9], [39].

C. MODELING AND CONTROL OF THE WT GENERATOR

A standard full-scale PMSG-based WT generator incorporating the mechanical dynamics of a two-mass gearbox system [1], [44] is used, as shown in Fig. 1. The output power of the WT-PMSG generator (P_{wt}), is expressed in (12a)–(12d) [9]

$$P_{wt} = 0.5 C_P (\lambda, \beta) \rho \pi R^2 v_{wt}^3 \quad (12a)$$

$$C_P (\lambda, \beta) = \frac{0.5176 (116 / \lambda_i - 0.4 \beta - 5)}{\exp \{21 / \lambda_i\}} + 0.0068 \lambda \quad (12b)$$

$$\lambda_i = \frac{1}{1 / (\lambda + 0.08 \beta) - 0.035 / (1 + \beta^3)} \quad (12c)$$

$$\lambda = \frac{R \omega_{wt}}{v_{wt}} \quad (12d)$$

where $C_P (\lambda, \beta)$ is the power coefficient of the WT generator, which exhibits a nonlinear characteristic across both PPP and off-PPP regions, i.e., under-rated-speed (URS) and over-rated-speed (ORS) regions, as illustrated in Fig. 3(b) [9].

The analytical equations governing the dynamics of the two-mass gearbox are addressed in (13a)–(13c) [1], [44]

$$2H_{wt} \frac{d}{dt} (\omega_{wt}) = T_{wt} - T_{sh} \quad (13a)$$

$$\frac{d}{dt} (\theta_{sh}) = \omega_b (\omega_{wt} - \omega_r^{\text{mech}}) \quad (13b)$$

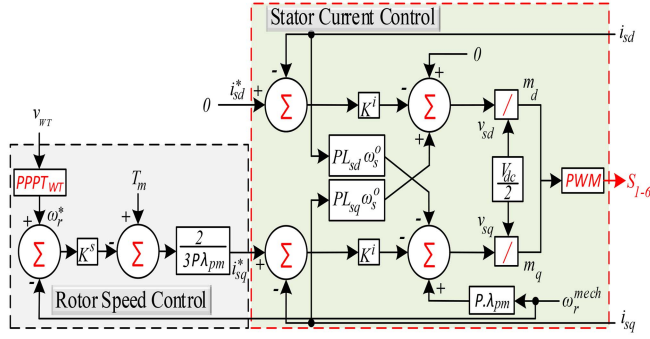


FIGURE 4. Schematic diagram of the VSR vector control for the WT-side circuit.

$$T_{sh} = K_{sh}\theta_{sh} + D_{sh}(\omega_{WT} - \omega_r^{mech}). \quad (13c)$$

The equations describing the dynamics of the PMSG's equivalent circuit model, as depicted in Fig. 1, are given in (14)–(15) [4]

$$\frac{d}{dt}(\phi_{dq}) + jP\omega_s^{\text{magn}}\phi_{dq} - R_s i_{sdq} = v_{sdq} \quad (14)$$

$$J_r \frac{d}{dt}(\omega_r^{mech}) + B_r \omega_r^{mech} = T_m - T_e \quad (15)$$

where ϕ_{dq} is the flux linkage of the PMSG in the dq -reference frame; ϕ_s and ϕ_r are the flux linkage of the stator and rotor magnets, respectively; and $\phi_d = \phi_{sd} + \phi_r$, $\phi_q = \phi_{sq}$, $\phi_{sdq} = -L_{sd}i_{sd} - jL_{sq}i_{sq}$, $\phi_r = \lambda_{pm}$, $\omega_s^{\text{magn}} = \omega_r^{mech}$, $T_m = T_{sh}$, $e_{sd} = P\omega_s^{\text{magn}}L_{sq}i_{sq}$, $e_{sq} = P\omega_s^{\text{magn}}(\lambda_{pm} - L_{sd}i_{sd})$, and $T_e = \text{Re}\{1.5e_{sdq}i_{sdq}^*\} = 1.5P\lambda_{pm}i_{sq}$ for surface-mounted PMSG, i.e., $L_{sd} = L_{sq}$.

The PPPT_{WT} block in Fig. 4 determines the optimal rotor speed value that results in the maximum power output of the WT generator [9]. To regulate the rotor speed (ω_r^{mech}) to the reference value (ω_r^{mech*}) and generate the q -component of the reference stator current (i_{sq}^*), a speed controller $K^s = k_p^s + k_i^s/s$ is employed in (16) while simultaneously setting i_{sd}^* to zero. These adjustments aim to maximize the generated torque (T), which is represented by the cross-product of the rotor flux (ϕ_r) and the stator flux (ϕ_s), i.e., $T = K(\phi_r \times \phi_s)$ [38]

$$i_{sq}^* = \frac{\eta_1 T_m - ((\omega_r^{mech})^* - \omega_r^{mech})(k_p^s + k_i^s/s)}{1.5P\lambda_{pm}}. \quad (16)$$

To mitigate the impact of the nonlinearity arising from the WT characteristic, a constant (η_1) is used in (16) to facilitate feedforward torque compensation, i.e., $T_m = P_m/\omega_r^{mech}$, which enables a rapid response of the rotor speed [9].

To regulate the stator currents (i_{sdq}) to their reference values (i_{sdq}^*), as illustrated in Fig. 4, and generate the modulation signals, a current controller $K^i = k_p^i + k_i^i/s$ is employed in (17). The conventional PWM produces the switching function

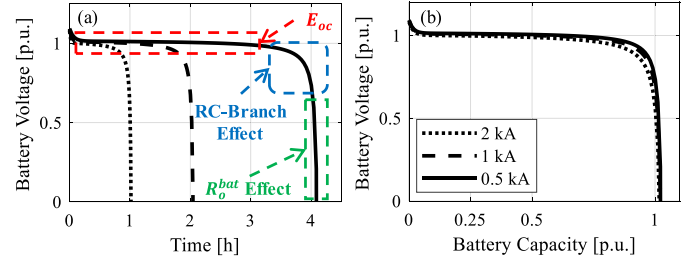


FIGURE 5. Characteristics of the battery at different discharging currents. (a) Voltage-time. (b) Voltage-battery capacity.

for the VSR

$$v_{sdq} = -(i_{sdq}^* - i_{sdq})\left(k_p^i + \frac{k_i^i}{s}\right) + jP\omega_s^{\text{magn}}\phi_{dq}. \quad (17)$$

D. MODELING AND CONTROL OF THE BESS

The configuration of the BESS system in Fig. 1 comprises a battery cabinet housing a 2-Q bidirectional converter designed to oversee charging and discharging processes [12]. The battery cabinet houses interconnected battery racks, each with multiple interconnected battery modules [10]. Each battery module, in turn, comprises numerous battery cells connected in series and parallel to meet the capacity, voltage, and current demands.

Numerous equivalent circuit models are presented in the literature to replicate batteries' internal characteristics and behavior, particularly during discharge. In this research study, the widely-adopted practical model illustrated in Fig. 1, which omits the self-discharging components, i.e., self-discharging impedance and self-discharging voltage, is employed, balancing simplicity and relatively accurate performance. The utilized equivalent circuit consists of three main components. First, an open circuit potential (E_{oc}) that symbolizes the battery's voltage. This potential is influenced by both temperature and the battery's state of charge (SoC). Second, an internal resistance (R_o^{bat}) that accounts for the instantaneous reaction originating from the electrolyte's resistance and the battery's internal structure. Finally, in the battery model shown in Fig. 1, a single RC parallel branch—comprising R_1^{bat} and C_1^{bat} —has been included to capture the battery's time-dependent tendencies and delayed response due to electrochemical processes. Other branches have been neglected to achieve a tradeoff between model complexity and accuracy. The dynamic behavior of this model is shown in Fig. 5 and mathematically represented in (18) and (19) [4], [12]

$$E_{oc} - V_{\text{bat}} = R_o^{\text{bat}}I_{\text{bat}} + R_1^{\text{bat}}I_{R_1^{\text{bat}}} \quad (18)$$

$$I_{\text{bat}} - I_{R_1^{\text{bat}}} = C_1^{\text{bat}}R_1^{\text{bat}}\frac{d}{dt}\left(I_{R_1^{\text{bat}}}\right). \quad (19)$$

To precisely assess the dynamic performance of the grid-tied hybrid system, it is crucial to examine the dynamic resistance of the battery during both the charging and discharging phases. The battery's resistance, as illustrated in

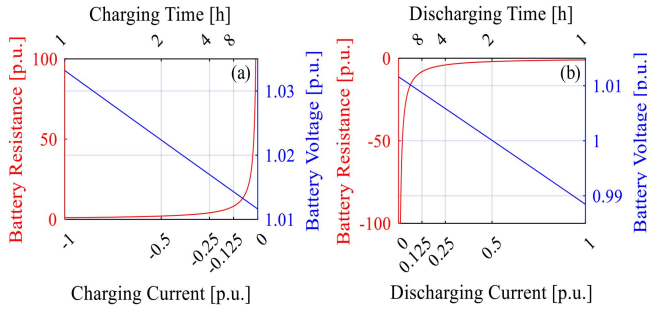


FIGURE 6. Battery resistance-current characteristics at different times. (a) Battery charging. (b) Battery discharging.

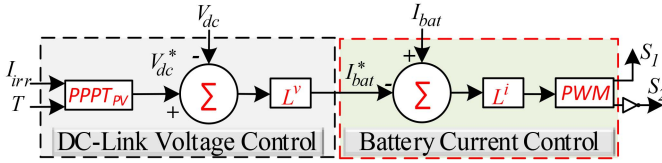


FIGURE 7. Schematic diagram of the 2-Q converter control for the BESS-side circuit.

Fig. 6, is determined by the V_{bat} to I_{bat} ratio. The resistance is significantly low, resulting in a negligible effect on the overall system performance, particularly at high current rates. A more detailed discussion regarding this aspect is given in Section III-B.

In Fig. 7, the PPPT_{PV} block is utilized to determine the optimal voltage resulting in the maximum power output from the PV generator and to produce the reference dc-link voltage (V_{dc}^*). To regulate the dc-link voltage (V_{dc}) and maintain it at the desired reference value, i.e., V_{dc}^* , the battery employs a dc-link voltage controller, $L^v = I_p^v + I_i^v/s$, which ensures that the dc-link voltage matches the reference optimal value and generates the corresponding reference battery current (I_{bat}^*), as described in the following equation:

$$I_{\text{bat}}^* = (V_{\text{dc}}^* - V_{\text{dc}}) \left(I_p^v + \frac{I_i^v}{s} \right). \quad (20)$$

A battery current controller, $L^i = I_p^i + I_i^i/s$, is utilized to regulate the battery current (I_{bat}) to match the reference value (I_{bat}^*) and generate the duty ratio ($\mathcal{D}$), as shown in (21) and Fig. 7. This duty ratio is then compared with a carrier wave to produce the complementary gating pulses S_1 and S_2 , which control the operation of the battery's 2-Q bidirectional converter [4]

$$\mathcal{D} = (I_{\text{bat}} - I_{\text{bat}}^*) \left(I_p^i + \frac{I_i^i}{s} \right). \quad (21)$$

As shown in Fig. 1, the battery and the 2-Q bidirectional converter operate synchronously in three modes based on the gating pulse values [4]. 1) When both S_1 and S_2 are inactive, or when they operate to maintain the battery's state without transferring energy, i.e., active for the same duration, the battery operates in the standby mode with no power flow through

the 2-Q bidirectional converter; 2) when S_1 is active, i.e., pulsing with a duty ratio of $\mathcal{D} = V_{\text{bat}}/V_{\text{dc}}$, and S_2 is inactive, the 2-Q bidirectional converter functions as a buck converter, placing the battery in the Ch mode; and 3) when S_2 is active, i.e., pulsing with a duty ratio of $1 - \mathcal{D}$, and S_1 is inactivated, the 2-Q bidirectional converter operates as a boost converter, placing the battery in the DCh mode.

The equivalent dynamic model of the 2-Q bidirectional converter that describes the Ch and DCh modes is expressed in (22a)–(22c)

$$L_{\text{bat}} \frac{d}{dt} (I_{\text{bat}}) + R_{\text{bat}} I_{\text{bat}} = V_{\text{bat}} - \mathcal{D} (V_{\text{dc}}) \quad (22a)$$

$$V_{\text{bat}} = P_{\text{bat}}/I_{\text{bat}} = -r_{\text{bat}} I_{\text{bat}} \quad (22b)$$

$$r_{\text{bat}} = R_o^{\text{bat}} + R_l^{\text{bat}} \parallel \frac{1}{sC_1^{\text{bat}}} + \dots + R_n^{\text{bat}} \parallel \frac{1}{sC_n^{\text{bat}}} \quad (22c)$$

where R_{bat} and L_{bat} are the resistance and inductance of the 2-Q bidirectional converter, respectively, and V_{bat} is the battery voltage expressed as $V_{\text{bat}} = P_{\text{bat}}/I_{\text{bat}}$, and can also be expressed as $V_{\text{bat}} = -r_{\text{bat}} I_{\text{bat}}$ to study the impact of the battery's dynamic resistance, where r_{bat} is the equivalent dynamic resistance of the battery.

The parameters of the system in Fig. 1 are typical [4], [9], [39], and they are given in Table 2.

E. POWER MANAGEMENT AT THE DC-LINK

As illustrated in Fig. 1, the operating modes of the VSI, VSR, and 2-Q converters are governed by the control and power management criteria. These criteria depend on several factors, including the output power from the PV and WT generators, the battery's SoC ratio, the battery's power limits during both Ch and DCh modes and the grid demand [24], [29]. The system continuously monitors the battery's SoC and adjusts controller reference values based on the PV and WT outputs and the grid demand; this ensures consistent and balanced power flow within the hybrid system [45].

The external charging and discharging power flows ($P_{\text{bat}}^{\text{ext}}$) into and out of the BESS system at the dc-link are governed by the following equation [25]:

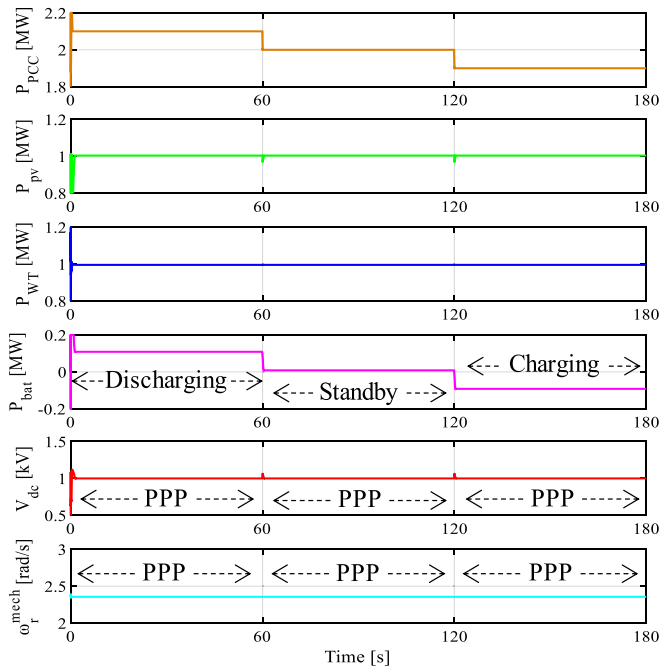
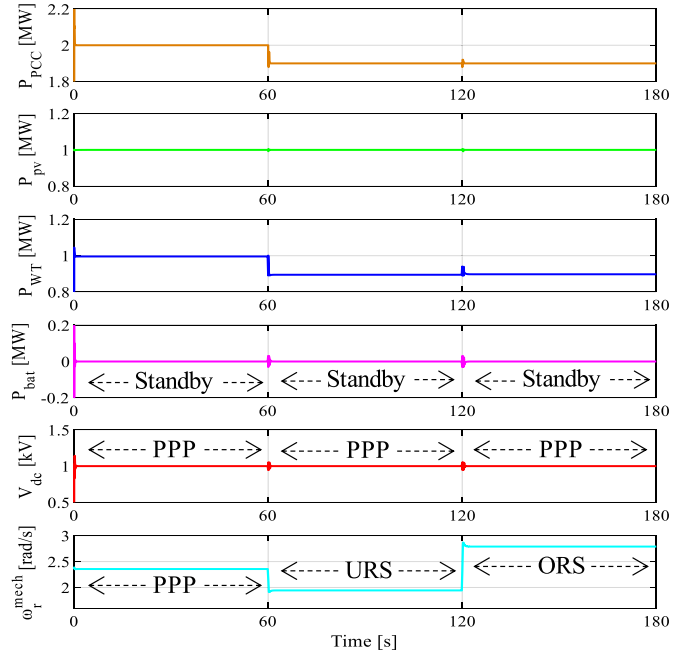
$$\begin{cases} C_{\text{dc}} V_{\text{dc}} \frac{d}{dt} (V_{\text{dc}}) = P_{\text{pv}}^{\text{ext}} + P_{\text{WT}}^{\text{ext}} \pm P_{\text{bat}}^{\text{ext}} - P_{\text{dc}} \\ C_{\text{dc}} = C_{\text{pv}} + C_{\text{WT}} + C_{\text{bat}} \\ P_{\text{pv}}^{\text{ext}} = I_{\text{pv}} V_{\text{dc}} \\ P_{\text{WT}}^{\text{ext}} \approx P_s = \mathcal{R}e \{ 1.5 v_s i_s^* \} = 1.5 (v_{s_d} i_{s_d} + v_{s_q} i_{s_q}) \\ P_{\text{bat}}^{\text{ext}} = [\mathcal{D} (I_{\text{bat}})] V_{\text{dc}} \\ P_{\text{dc}} \approx P_{\text{inv}} = \mathcal{R}e \{ 1.5 v_i i^* \} = 1.5 (v_{i_d} i_d + v_{i_q} i_q) \end{cases} \quad (23)$$

where C_{dc} is the parallel equivalent dc-link capacitor, consisting of C_{pv} , C_{WT} , and C_{bat} .

Figs. 8–10 illustrate the power management process and the total power demand that needs to be supplied to the grid through the dc-link. The PV and WT generators can be optimized to operate within their respective PPP regions, collectively generating 2.0 MW or below when operating under their respective off-PPP regions. As detailed below, the 1.0 MWh BESS comes into play in four operational cases,

TABLE 2. Parameters of the Nonoscillatory Highly Damped Weak Grid-Tied Hybrid PV-WT-BESS System

Parameter	Value / Unit	Parameter	Value / Unit
<i>DC-Link Grid-Side</i>			
e_g	12.47 kV	R_{sh}	60.18 Ω
ω^o	$2\pi(60 \text{ Hz})$	α_i	$0.05 \%^\circ\text{C}^{-1}$
SCR	2	I_{irr}^*	1000 Wm^{-2}
X/R	7	T^*	298.15 K
N	0.48 kV/12.47 kV	E_g	0.1762 aJ
L_p	122.23 μH	k_B	$13.81 \text{ yJ}^\circ\text{K}^{-1}$
R_p	4.61 m Ω	<i>DC-Link WT-Side</i>	
L_s	27.5 mH	C_{WT}	5000 μF
R_s	1.04 Ω	R_s	0.82 m Ω
L_f	100 μH	L_{sd}	1.57 mH
R_f	2 m Ω	L_{sq}	1.57 mH
C_f	369 μF	P	26
f_{sw}	51(60 Hz)	λ_{pm}	5.83 Wb
r_{on}	1 m Ω	J_r	32170 kgm^2
g_p^δ	$90 \text{ V}^{-1}\text{s}^{-1}$	H_{WT}	0.1 s
g_i^δ	$5 \text{ kV}^{-1}\text{s}^{-2}$	ω_b	0.375 Hz
g_p^i	0.19 Ω	K_{sh}	1 p.u.
g_i^i	$3.85 \Omega\text{s}^{-1}$	D_{sh}	1 p.u.
g_p^v	100 VArV^{-1}	f_{sw}	51(60 Hz)
g_i^v	$500 \text{ kVArV}^{-1}\text{s}^{-1}$	k_p^i	3.02 Ω
<i>DC-Link PV-Side</i>		k_i^i	$1.58 \Omega\text{s}^{-1}$
C_{PV}	5000 μF	k_p^s	610 kNm
L_{dc}	0.34 μH	k_i^s	1320 kNm
R_{dc}	0.125 m Ω	<i>DC-Link BESS-Side</i>	
N_p	59	C_{bat}	5000 μF
N_s	29	R_{bat}	0.1 Ω
n_s	120	L_{bat}	0.5 mH
A	0.47	E_{oc}	0.5 kV
I_{sc}	18.37 A	f_{sw}	20 kHz
I_{rs}^*	10.43 pA	l_p^i	0.005 A^{-1}
q	0.16 aC	l_i^i	$3 \text{ A}^{-1}\text{s}^{-1}$
R_s	0.10 Ω	l_p^v	$0.45 \Omega^{-1}$
		l_i^v	$15 \Omega^{-1}\text{s}^{-1}$

**FIGURE 8.** Power management under varying grid demands and SoC < 100% .**FIGURE 9.** Power management under varying grid demands, different rotor speed levels, and SoC = 100% .

depending on the battery's SoC level and the grid's energy demand.

Case 1: During the time interval from 0 to 60 s, as shown in Fig. 8, the BESS supplements the maximum output power of 2 MW generated by the PV and WT systems to meet grid demand when it exceeds 2 MW.

Case 2: During the time interval from 60 to 120 s, as illustrated in Fig. 8, the BESS remains on standby, as the grid demand is exactly fulfilled at 2 MW, matching the maximum combined output of the PV and WT systems.

Case 3: During the time interval from 120 to 180 s, as depicted in Fig. 8, the BESS absorbs excess power generated by the PV and WT systems when grid demand falls below 2 MW, provided the battery's SoC level is below 100% .

Case 4: When the BESS is fully charged, i.e., SoC = 100%, any excess power generated beyond grid demand must be efficiently managed to prevent power wastage and maintain system stability [4]. Such management may involve discharging surplus power into additional standby storage systems, supplying the grid demands of neighboring regions, or dissipating excess energy through heat conversion using a tiebreaker load. Alternatively, curtailment strategies could be employed. Such curtailment management may entail modifying rotor speed settings, as depicted in Fig. 9, or adjusting the dc-link voltage, as shown in Fig. 10, to operate in the off-PPP regions of the WT and PV generators, respectively. However, it's important to note that adjusting the dc-link voltage, rather than rotor speed, might affect system stability (see Fig. 10), as will be further elaborated in Section III-B. In Fig. 10, the battery operates in the standby mode, whereas the PV and WT operate in the CLR and PPP, respectively, to meet the grid's

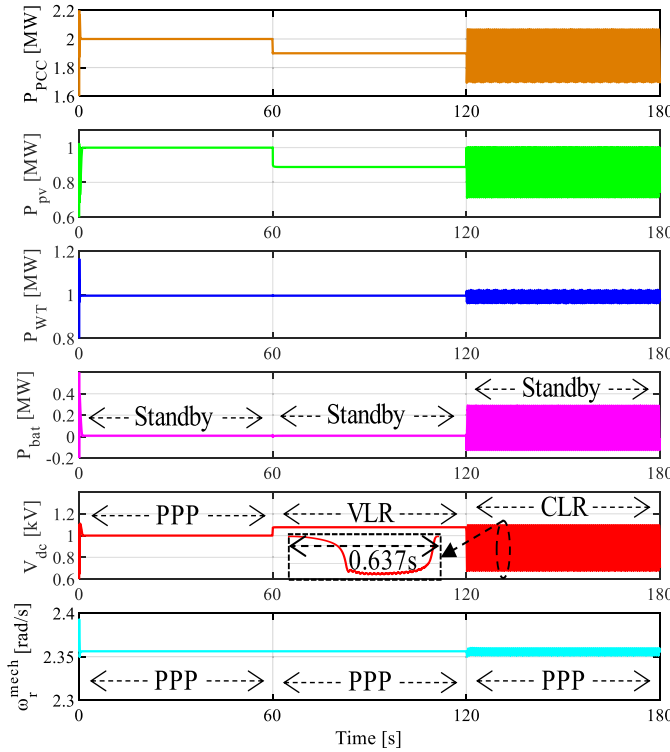


FIGURE 10. Power management under varying grid demands, different PV voltage levels, and SoC = 100% .

energy demand. However, the system became unstable due to the interaction dynamics between the highly dynamic resistance of the PV system in the CLR region and the high grid impedance. This adverse interaction caused all energy sources to oscillate across different operating regions unstably, preventing them from maintaining their intended operating states, as illustrated in Fig. 10.

III. PV-WT-BESS DYNAMIC INTERACTIONS AT WEAK GRID CONDITIONS

An LSSM model is developed by applying perturbations to the TDNLM model's differential equations $\dot{x} = F(x(t), u(t), t)$ in (1)–(23). The resulting LSSM model comprises 24 states and is provided in Appendix A.

In this section, the developed LSSM model is validated by comparing it to the real-time and offline TDNLM models, and the dynamic stability of the overall system is examined under weak grid conditions and various operating scenarios.

A. VERIFICATION OF THE DEVELOPED LSSM MODEL

The accuracy of the developed LSSM model with the initial operating points within the stable regions is evaluated for the four scenarios depicted in Figs. 8–10. Fig. 11 shows the results for the nonoscillatory highly damped design conditions described in Table 2. Moreover, Fig. 12 evaluates the accuracy for oscillatory lightly damped design conditions described in Table 3. The assessment involves applying a 5% step increase to the nonoscillatory highly damped P_{pcc} response in

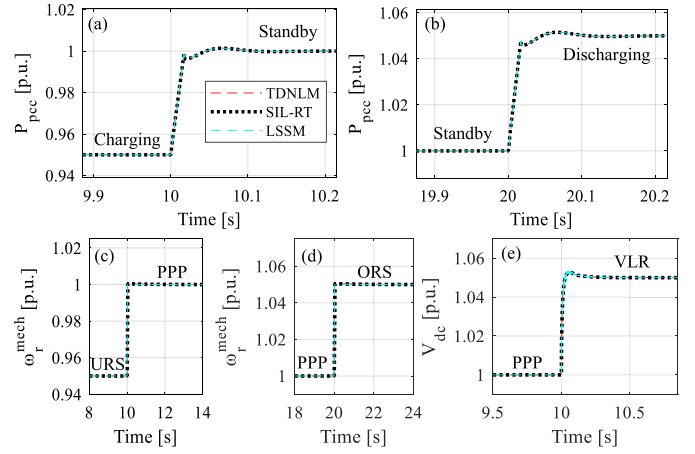


FIGURE 11. Verification of the LSSM model under nonoscillatory highly damped conditions. (a) and (b) PCC demand power. (c) and (d) Rotor mechanical speed step responses. (e) DC-link voltage step response.

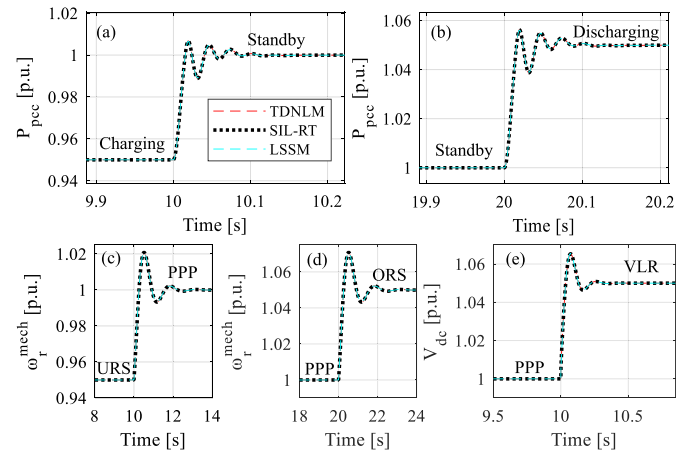


FIGURE 12. Verification of the LSSM model under oscillatory lightly damped conditions. (a) and (b) PCC demand power. (c) and (d) Rotor mechanical speed step responses. (e) DC-link voltage step response.

TABLE 3. Parameters of the Oscillatory Lightly Damped Weak Grid-Tied Hybrid PV-WT-BESS System

Parameter	Value / Unit
g_p^i	0.02 Ω
k_p^s	110 kNms
l_p^v	0.25 Ω^{-1}

Fig. 11(a)–(b), the response of ω_r^{mech} in Fig. 11(c) and (d), and the response of V_{dc} in Fig. 11(e). The results are then compared with the real-time and offline TDNLM models. In addition, oscillatory lightly damped responses are intentionally induced by specifically modifying the parameters g_p^i , k_p^s , and l_p^v , as provided in Table 3, consequently reducing the phase margins of the VSI current controller in Fig. 12(a) and (b), the rotor speed controller in Fig. 12(c) and (d), and the dc-link voltage controller in Fig. 12(e). The intentionally oscillatory lightly damped adjustment is performed exclusively in this section to demonstrate the verification between the

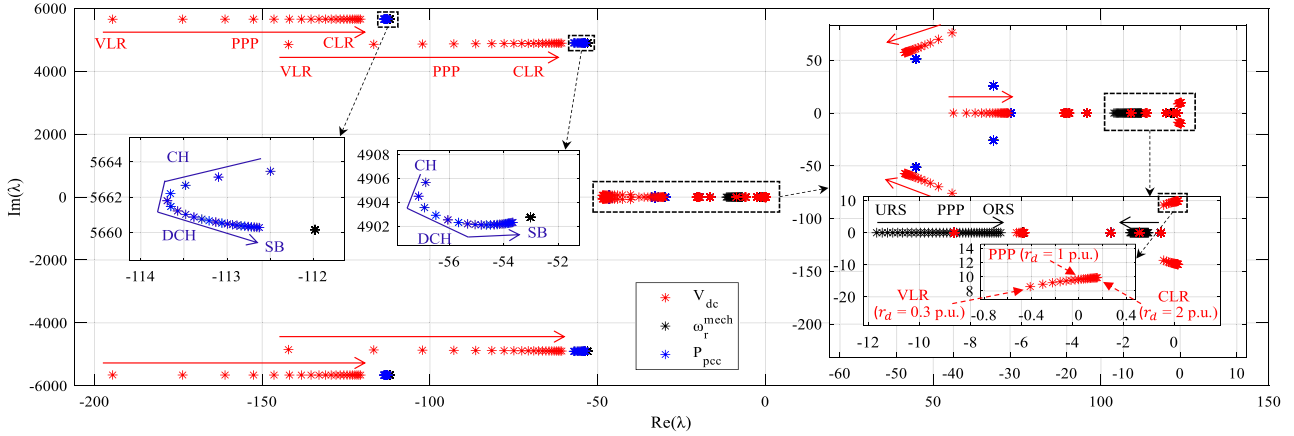


FIGURE 13. Eigenvalues analysis of the undamped LSSM model as the status of BESS changes from Charging to Standby through Discharging, the WT operating points shift from URS to ORS through PPP, and the PV operating points change from VLR to CLR through PPP.

LSSM model and the real-time and offline TDNLM models for P_{pcc} , ω_r^{mech} , and V_{dc} , respectively, as illustrated in Fig. 12.

To further characterize the validity of the LSSM model, it is essential to note that small-signal linearization remains accurate for disturbances up to approximately 5%–10% of the nominal operating point, as typically observed in power and energy systems. For disturbances exceeding 10%–15%, nonlinear effects become increasingly significant, leading to deviations between the LSSM and TDNLM results. Beyond 20%, the system behavior becomes strongly nonlinear, requiring nonlinear-model analysis or simulation tools for enhanced accuracy. This threshold aligns with classical control theory, where linearized models are reliable within a limited range around the equilibrium point, beyond which higher-order nonlinearities must be considered.

B. DYNAMIC RESISTANCE OF ENERGY SOURCES AND INTERACTION DYNAMICS WITH WEAK GRID

The operating points of both PV and WT generators exhibit fluctuations between PPP and off-PPP regions, influenced by rapidly changing atmospheric conditions [9], [15]. Meanwhile, the battery's operating points change between Ch and DCh modes, influenced by the grid demand [4], [45]. This section investigates the impact of the dynamic resistance on the dc-side of the overall weak grid-tied system, i.e., $SCR = 2$. The study explored all battery operating regions and PPP and off-PPP regions of PV and WT generators. It should be noted that the nonoscillatory highly damped controllers design is considered for the LSSM and offline TDNLM models discussed in Section III-A. The complete LSSM model of the undamped system is provided in Appendix B.

1) INTERACTION DYNAMICS OF THE UNDAMPED SYSTEM-BASED DIFFERENT BESS OPERATING REGIONS

From the dynamic stability perspective, the battery's dynamic resistance exhibits no significant effect. It remains negligible at high charging and discharging current levels, as depicted

in Fig. 6(a) and (b). The eigenmode stability analysis of the entire system is illustrated in Fig. 13. As the battery's operating point-dependent dynamic resistance transitions between Ch, SB, and DCh modes, it is evident that the eigenmodes migrate within the left-half-plane (LHP) of the S-plane without violating the stability margins.

Fig. 14(a) demonstrates that the open-loop frequency response may exhibit either negative/positive phase margins (–ve/+ve PM) or negative/positive gain margins (–ve/+ve GM), indicating a potential instability. Bode plots become inconclusive for systems with potential right-half plane (RHP) singularities. Therefore, a more conservative approach, the generalized Nyquist stability criterion based on Cauchy's argument principle $Z = N + P$ [46], is adopted to analyze the system's stability comprehensively. In this criterion, N represents the number of resultant contours encircling the point $(-1 + j0)$ in the clockwise direction, P is the number of poles of the open-loop transfer function in the RHP, and Z is the number of poles of the closed loop system in the RHP.

In Fig. 14(a), the open-loop dc-link voltage dynamics exhibit $P = 2$, $P = 2$, and $P = 1$ RHP poles across charging, discharging, and standby battery operating regions, respectively. The clockwise encirclements of the open-loop system in Fig. 14(b) are $N = -2$, $N = -2$, and $N = -1$ across charging, discharging, and standby battery operating regions, respectively. As a result, the entire closed-loop system in Fig. 14(c) and Table 4 demonstrates stable performance across all battery operating conditions, with zero RHP poles, i.e., $Z = 0$ under the Ch, DCh, and SB regions.

2) INTERACTION DYNAMICS OF THE UNDAMPED SYSTEM-BASED DIFFERENT WT OPERATING REGIONS

As the WT's operating point-dependent dynamic resistance, influenced by varying rotor speed reference levels, transitions from the URS to the ORS through the PPP, the machine-side eigenmodes in Fig. 13 migrate within the LHP of the S-plane. Notably, the remaining eigenmodes remain consistent without violating stability margins.

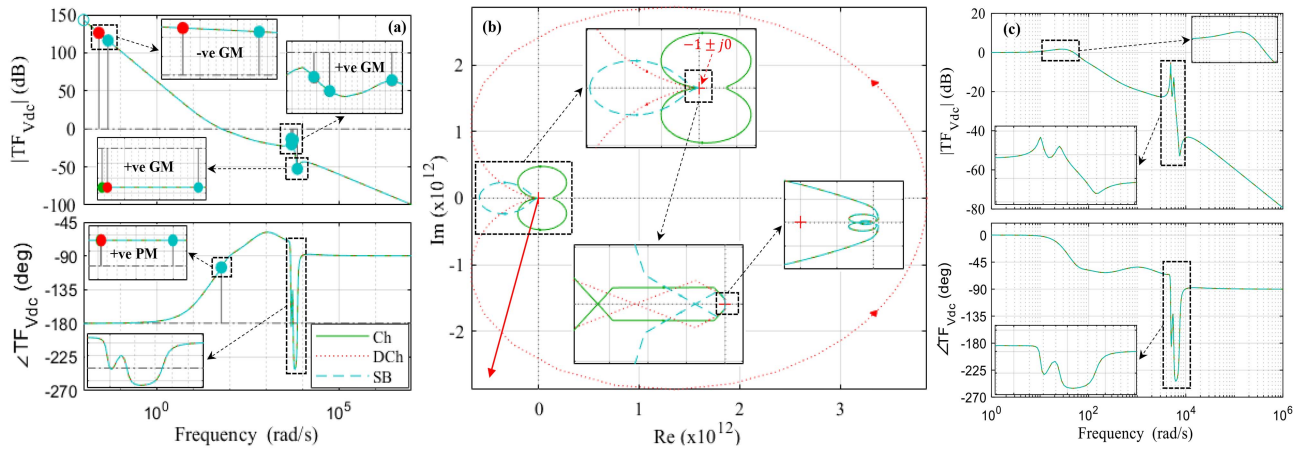


FIGURE 14. Undamped system as the status of BESS changes from charging to standby through discharging. (a) Bode plot of the open-loop dc-link voltage dynamics. (b) Nyquist analysis of the open-loop dc-link voltage dynamics. (c) Bode plot of the closed-loop dc-link voltage dynamics.

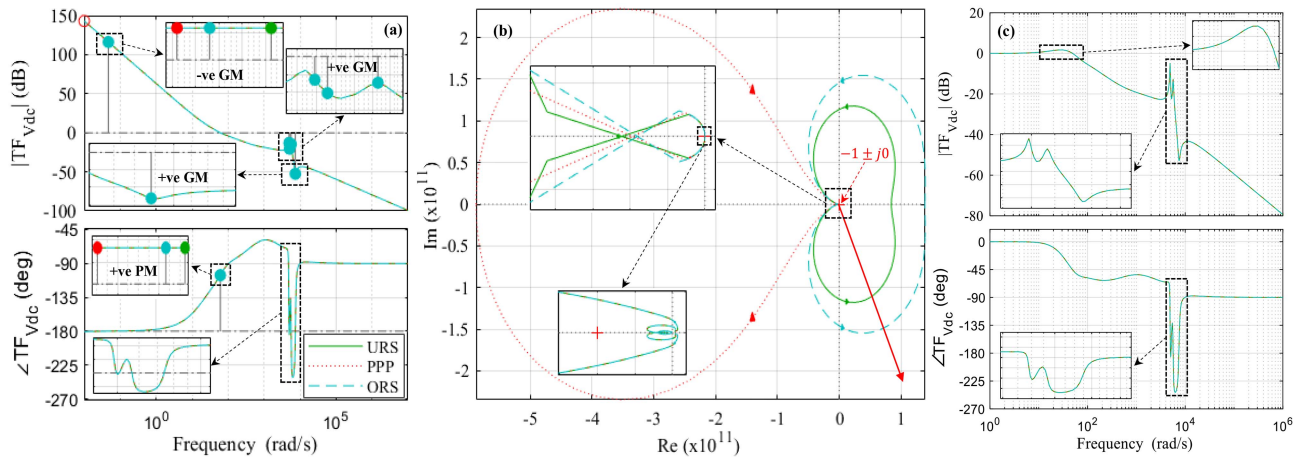


FIGURE 15. Undamped system as the WT operating points shift from URS to ORS through PPP. (a) Bode plot of the open-loop DC-link voltage dynamics. (b) Nyquist analysis of the open-loop DC-link voltage dynamics. (c) Bode plot of the closed-loop DC-link voltage dynamics.

TABLE 4. Nyquist Stability Criterion Based on Cauchy's Argument Principle for Undamped and Damped Models Under Different Operating Regions

Nyquist stability criterion based on Cauchy's argument principle						
Model	Operating Region		N	P	Z = N + P	Stability
Undamped	BESS	Ch	-2	2	0	Stable
		SB	-1	1	0	Stable
		DCh	-2	2	0	Stable
	PV	VLR	-2	2	0	Stable
		PPP	-2	2	0	Stable
		CLR	2	0	2	Unstable
	WT	URS	-2	2	0	Stable
		PPP	-1	1	0	Stable
		ORS	-2	2	0	Stable
Damped	PV	VLR	-1	1	0	Stable
		PPP	-2	2	0	Stable
		CLR	0	0	0	Stable

In Fig. 15(a), the open-loop frequency response of the dc-link dynamics indicates potential negative phase or gain margins, necessitating a more rigorous stability assessment through Nyquist analysis. The Nyquist analysis is conducted

and shown in Fig. 15(b). It accurately examines the system's dynamic stability under different WT operating regions. The open-loop dc-link voltage dynamics, shown in Fig. 15(a), exhibit RHP poles of $P = 2$, $P = 1$, and $P = 2$ across the URS, PPP, and ORS operating regions, respectively. In addition, Fig. 15(b) demonstrates that the system exhibits clockwise encirclements of $N = -2$, $N = -1$, and $N = -2$ in the URS, PPP, and ORS regions, respectively. As a result, Fig. 15(c) and Table 4 confirm that the system's stability remains unaffected by changes in machine-side operating points and is stable throughout all WT operating regions, with zero RHP poles, i.e., $Z = 0$ in the URS, PPP, and ORS regions.

3) INTERACTION DYNAMICS OF THE UNDAMPED SYSTEM-BASED DIFFERENT PV OPERATING REGIONS

The dynamic resistance of the PV system varies significantly across different regions. The PV dynamic resistance in the VLR and PPP regions is minimal, contrasting with its relatively large value in the CLR region, as illustrated in Fig. 3(a). This higher resistance in the CLR region can adversely

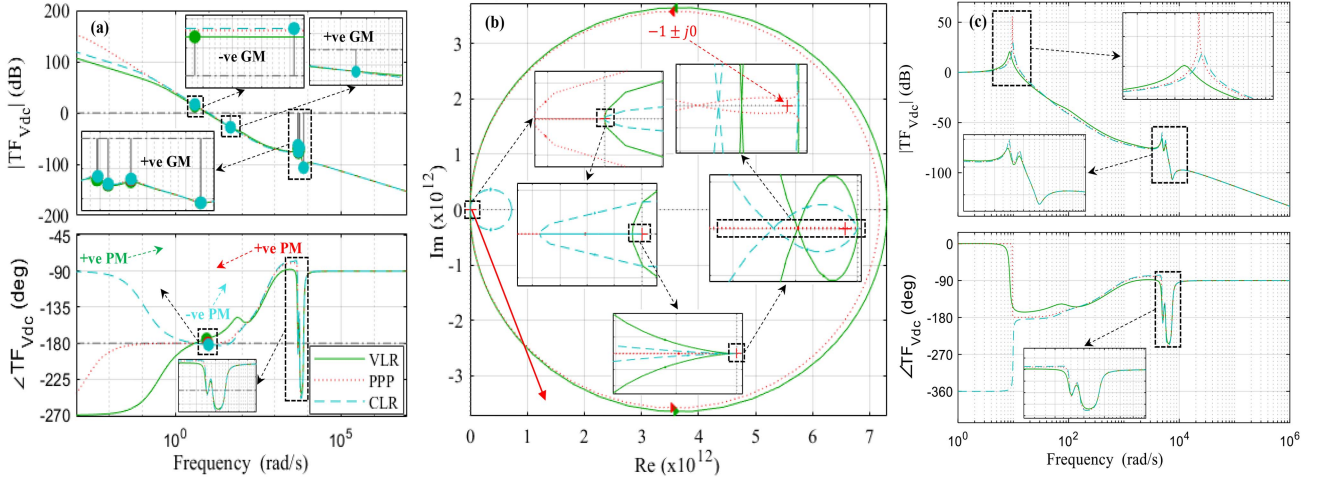


FIGURE 16. Undamped system as the PV operating points change from VLR to CLR through PPP. (a) Bode plot of the open-loop DC-link voltage dynamics. (b) Nyquist analysis of the open-loop DC-link voltage dynamics. (c) Bode plot of the closed-loop DC-link voltage dynamics.

impact the system's stability, particularly under weak grid conditions.

As the operating point-dependent PV dynamic resistance, influenced by changes in the reference dc-link voltage level, transitions from the VLR to PPP, as shown in Fig. 13, the eigenmodes migrate within the LHP of the S-plane without breaching stability margins. However, when moving from the PPP to the CLR, a critical shift occurs, leading to eigenmodes migrating to the RHP, compromising the system's stability.

In Fig. 16(a), the open-loop dc-link voltage frequency response reveals negative phase and gain margins across all PV operating regions. Consequently, the Nyquist criterion is employed in Fig. 16(b) to assess the system's dynamic stability precisely. The transfer function of the open-loop dc-link dynamics in Fig. 16(a) exhibits RHP poles of $P = 2$, $P = 2$, and $P = 0$ across the VLR, PPP, and CLR, respectively. The clockwise encirclements in Fig. 16(b) amount to $N = -2$, $N = -2$, and $N = 2$ for the same regions. Therefore, as shown in Fig. 16(c) and Table 4, the closed-loop dc-link voltage dynamics are stable in the VLR and PPP regions without RHP poles, i.e., $Z = 0$. However, in the CLR region, the system becomes entirely unstable with two RHP poles, i.e., $Z = 2$, indicating the presence of two low-frequency conjugate poles at 9.86 rad/s.

IV. PROPOSED ACTIVE STABILIZATION METHOD

The preceding section demonstrated that the hybrid system could encounter LFO instabilities under SoC = 100%, SCR = 2, and CLR operating region conditions. This section proposes an effective active damping method to mitigate these undesirable dynamic interactions. This method involves conducting a participation factor analysis to explore the sensitivity of the unstable eigenmodes to specific state variables. The proposed active damping technique is then applied to stabilize the entire system under unstable conditions. In addition, an analysis of both undamped and damped systems is conducted using the developed ISAM model to demonstrate the effectiveness of

TABLE 5. Participation Factor Calculation for the Undamped System in Response to Changes in the PV Operating Points

Eigenmodes (λ)	State(s)	Eigenmodes (λ)	State(s)
$-6.7651E + 6$	i_{sq}	-19.9093	φ_d^L
$-1.0087E + 6$	φ^{G^v}, I_{pv}	-19.6713	φ_q^L
$-112.7772 \pm j5662.7705$	i_{gd}^g, v_d^g	-0.5219	ω_r^{mech}
$-50.2698 \pm j4899.0416$	i_{gq}^g, v_q^g	-1922.6547	$\varphi_d^{K^L}$
-2016.8833	i_d^g	-1922.6547	$\varphi_q^{K^L}$
-1631.0414	i_q^g	-16.4671	ω_{WT}
$-49.7310 \pm j57.1316$	δ^c, φ^{G^s}	-0.5219	φ^{K^s}
$0.7631 \pm j9.9365$	V_{dc}	-8.6357	I_{bat}
-4.6179	φ^{L^p}	-2.4949	θ_{sh}
-32.1016	i_{sd}	-1.3642	φ^{L^l}

the proposed stabilization technique for weak grid conditions, i.e., $1 < SCR < 2$, and different PV operating regions.

A. PARTICIPATION FACTOR ANALYSIS

A participation factor analysis is conducted on the unstable system model to identify the state variables that most influence the instability [9], [15], [40]. The participation factor (p_{ki}), which represents the relationship between the k th state variable and the i th eigenvalue, is defined as $p_{ki} = \partial \lambda_i / \partial a_{kk}$, where a_{kk} represents the diagonal elements of the system matrix. As shown in Table 5, the analysis reveals that the V_{dc} state has a dominant influence on the LFO unstable eigenmodes, $\lambda = 0.7631 \pm j9.9365$, with a participation factor of 0.99.

B. PROPOSED ACTIVE DAMPING TECHNIQUE

To mitigate undesirable LFO instabilities associated with the dynamic interactions in the CLR region and weak grid conditions, an active damping control technique is utilized due to its minimal power losses compared to passive damping methods, particularly in large-scale power systems [7], [9], [15]. As demonstrated in Section IV-A, it was established that the LFO unstable eigenmodes are primarily influenced by the V_{dc} state.

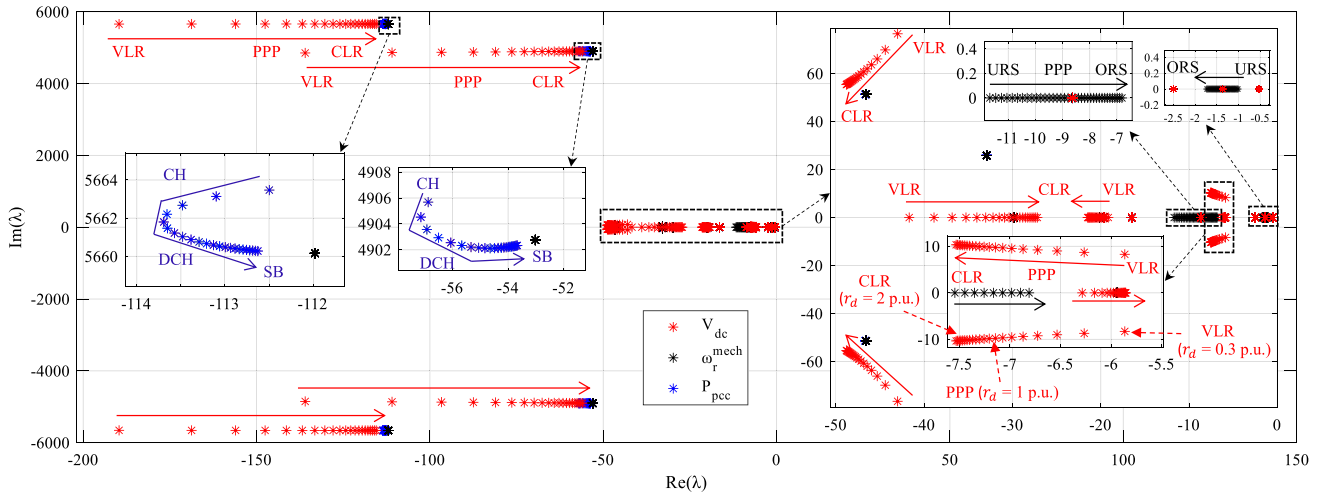


FIGURE 17. Eigenvalues analysis of the damped LSSM model as the status of BESS changes from Charging to Standby through Discharging, the WT operating points shift from URS to ORS through PPP, and the PV operating points change from VLR to CLR through PPP.

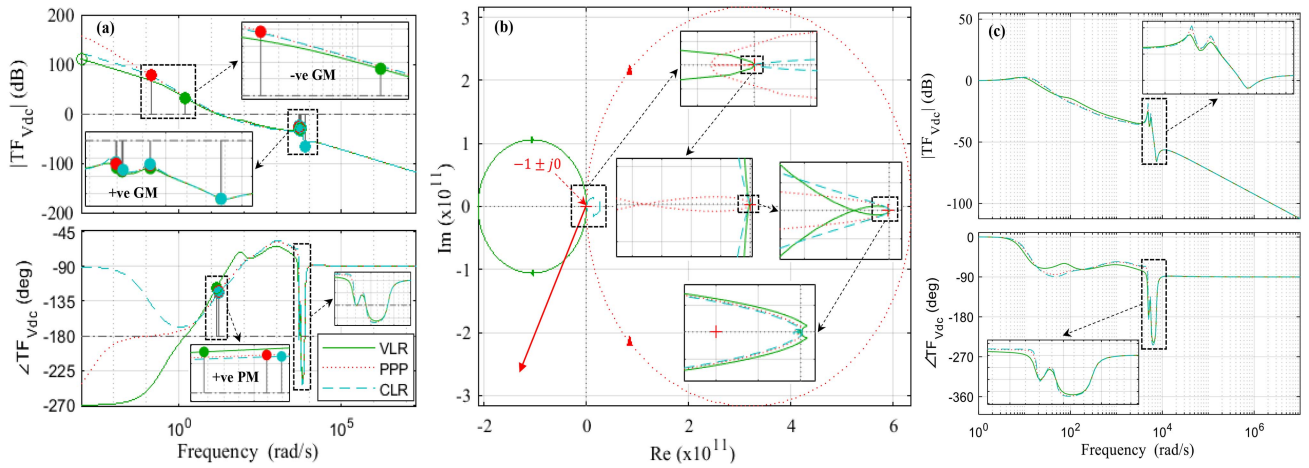


FIGURE 18. Damped system as the PV operating points change from VLR to CLR through PPP. (a) Bode plot of the open-loop DC-link voltage dynamics. (b) Nyquist analysis of the open-loop DC-link voltage dynamics. (c) Bode plot of the closed-loop DC-link voltage dynamics.

Consequently, as indicated in (24), an altered version of V_{dc} is introduced into the output of the dc-link voltage control loop (I_{bat}^*) to reshape the system transfer functions. This modification effectively eliminates the LFO unstable eigenmodes and actively improves the performance of the dc-link response

$$I_{bat}^*|_{new} = I_{bat}^* + I_{dmp}^{LFO}. \quad (24)$$

The proposed damping loop yields the damping current signal (I_{dmp}^{LFO}), given in the following equation:

$$I_{dmp}^{LFO} = \alpha [\tau s / (\tau s + 1)] V_{dc} \quad (25)$$

where $\alpha = -3$ is a constant chosen to enhance the LFO damping and $LCF(s) = \tau s / (\tau s + 1)$ is a low-cut filter with a time constant $\tau = 0.5$ ms, ensuring operation outside the filter's transition band region.

Figs. 17 and 18 illustrate the impact of the applied damping controller in (24)–(25) on the undamped system shown in Figs. 13 and 16, respectively, under different PV operating

TABLE 6. System Stability Performance Under Different PV Operating Regions and Severe Weak Grid Conditions

	Undamped Model	Damped Model
PV Region	Weak Grid (SCR=2)	
VLR	Stable	Stable
PPP	Stable	Stable
CLR	Unstable	Stable
Grid Stiffness	CLR Region	
SCR=2	Unstable	Stable
SCR=1.2	Unstable	Stable

regions, SoC = 100%, and weak grid conditions. As shown in Fig. 17, the proposed damping approach successfully relocates the unstable eigenmodes to the LHP when transitioning from the VLR to the CLR through the PPP while preserving system stability across all WT and BESS operating regions.

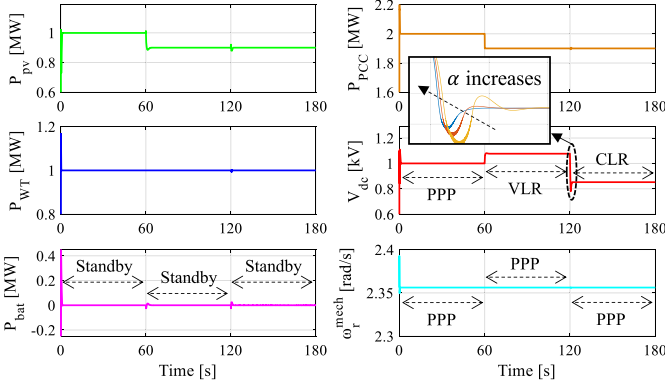


FIGURE 19. Damped system under various PV voltage levels and SoC = 100% .

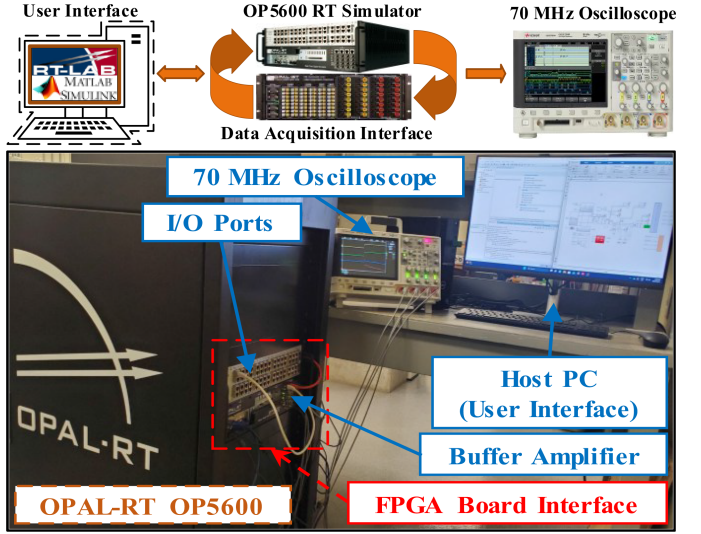


FIGURE 22. OPAL-RT real-time SIL test setup.

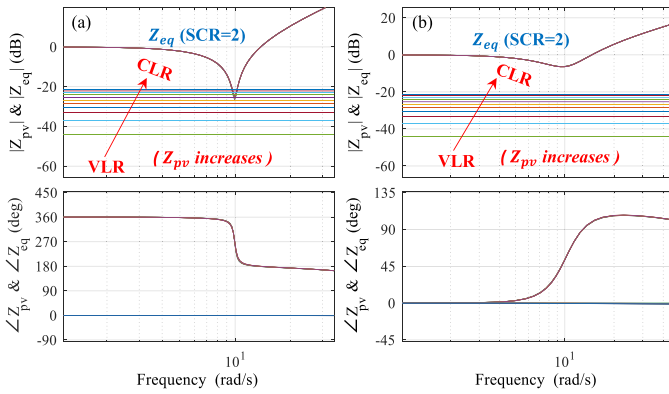


FIGURE 20. Small-signal impedances Z_{pv} and Z_{eq} under various PV operating regions and SCR = 2. (a) Undamped model. (b) Damped model.

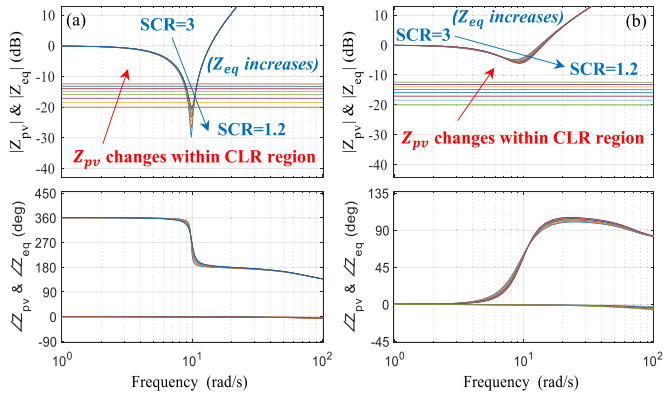


FIGURE 21. Small-signal impedances Z_{pv} and Z_{eq} at various grid stiffness and the CLR region. (a) Undamped model. (b) Damped model.

For the damped system depicted in Fig. 18, the open-loop dc-link voltage transfer function exhibits $P = 1$, $P = 2$, and $P = 0$ RHP poles [see Fig. 18(a)], with clockwise encirclements of $N = -1$, $N = -2$, and $N = 0$ [see Fig. 18(b)] across the VLR, PPP, and CLR regions, respectively. Consequently, as shown in Fig. 18(c) and Table 4, the closed-loop dc-link voltage dynamics are stable across all PV operating regions, with zero RHP poles, i.e., $Z = 0$ under VLR, PPP, and CLR regions.

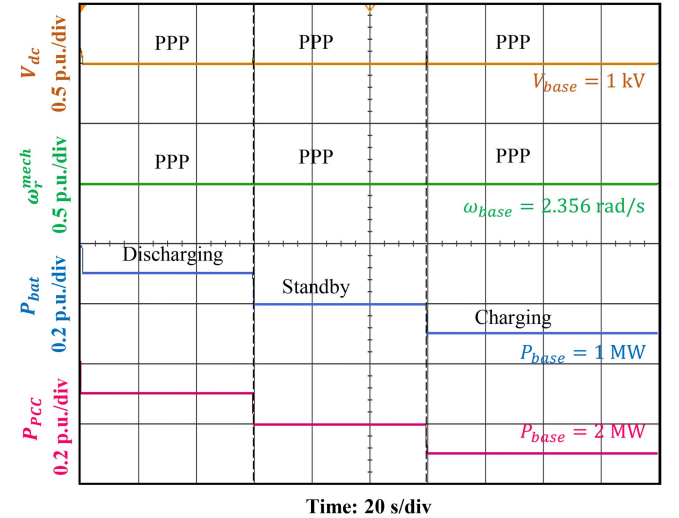


FIGURE 23. Power management at different grid demands and SoC < 100% .

Compared to Figs. 10, and 19 demonstrates that the proposed compensated system robustly preserves dynamic stability in the VLR and PPP regions while successfully eliminating the LFO instability in the CLR region by reshaping the dc-link voltage transfer function, as shown in Fig. 18(c). More importantly, the suggested stabilization approach has no impact on the system's steady-state performance, as the damping current signal is zero in steady-state conditions, i.e., $\lim_{t \rightarrow \infty} I_{dmp}^{LFO}(t) = \lim_{s \rightarrow 0} s I_{dmp}^{LFO}(s) = 0$. An additional state in (25) is introduced, and the complete LSSM model of the damped system is provided in Appendix C.

C. IMPEDANCE-BASED STABILITY ANALYSIS

In this section, an impedance-based stability modeling analysis is conducted to elaborate on the dc-side interaction

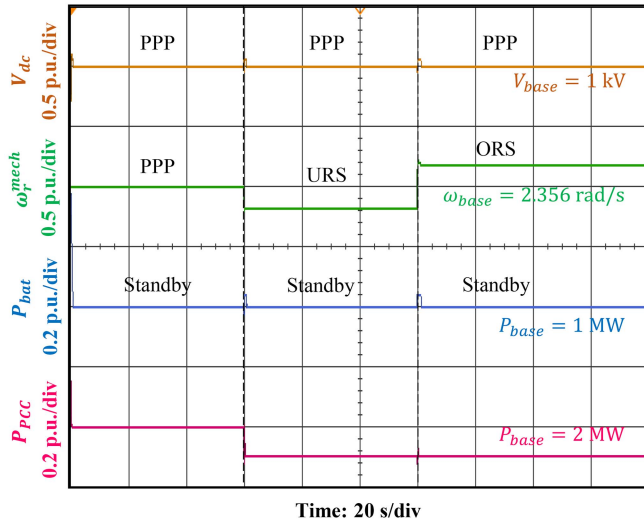


FIGURE 24. Power management at different grid demands, SoC = 100%, and various rotor speed levels.

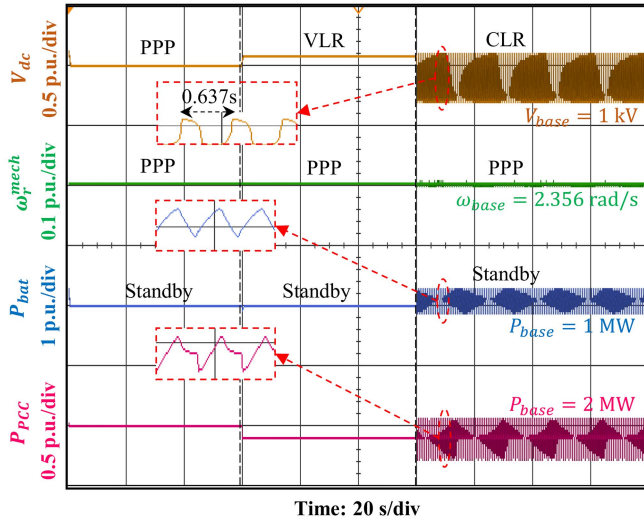


FIGURE 25. Power management of the undamped system under different grid demands, SoC = 100%, and various PV voltage levels.

dynamics between the dynamic impedance of the PV system and the equivalent impedance of the rest of the system units. As observed from the dc terminals, the small-signal impedance of the dc-link PV-side system, denoted as Z_{pv} , is given by (26). Meanwhile, the overall parallel small-signal equivalent dc-side impedance of the VSI (Z_{inv}), WT (Z_{WT}), and BESS (Z_{bat}), is derived from the state-space model and is given by (27).

$$\Delta Z_{pv} = \frac{1}{sC_{pv}} \parallel [(sL_{dc} + R_{dc}) + r_d] \quad (26)$$

$$\Delta Z_{eq} = \Delta Z_{inv} \parallel \Delta Z_{WT} \parallel \Delta Z_{bat}. \quad (27)$$

In Fig. 20 and Table 6, ΔZ_{pv} and ΔZ_{eq} are investigated at different PV operating regions, SCR = 2, and with/without active damping. Fig. 20(b) clearly illustrates that the

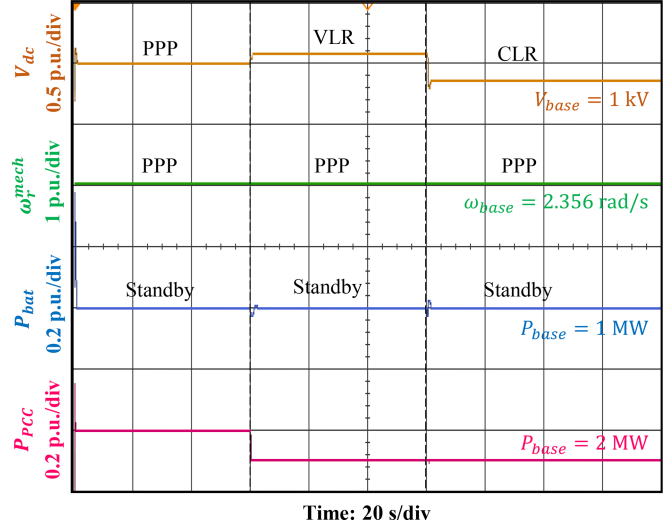


FIGURE 26. Power management of the damped system under different grid demands, SoC = 100%, and various PV voltage levels.

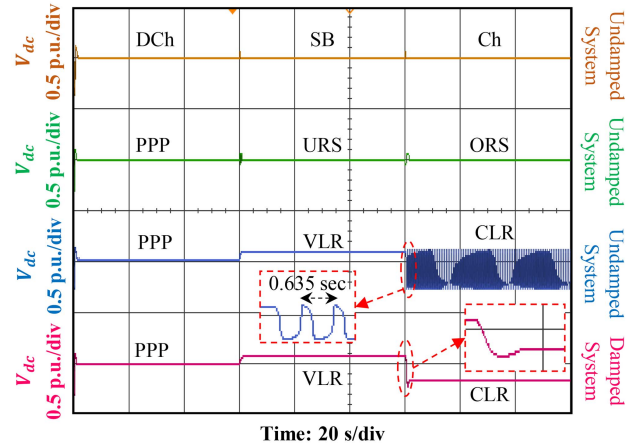


FIGURE 27. Performance of the undamped and damped systems under SCR = 1.2 and different operating regions.

low-frequency dynamic interactions in the damped system are mitigated across all PV operating regions, including the CLR region, compared to the undamped system in Fig. 20(a).

Similarly, Fig. 21 and Table 6 illustrate the variation in undamped and damped system impedances at the CLR operating region, with grid conditions ranging from weak grid operation, i.e., SCR=3, to severely weak grid operation, i.e., SCR=1.2. Fig. 21(b) emphasizes the effectiveness of the proposed damping technique, even under very low SCR values, and demonstrates that the stability of the damped system is preserved compared to the undamped system in Fig. 21(a).

V. REAL-TIME SIMULATION VALIDATION RESULTS

Real-time simulations, using the OPAL-RT OP5600 setup shown in Fig. 22, are used to validate the LFO instability in grid-tied hybrid PV-WT-BESS system and the effectiveness of the proposed active damping approach. Real-time

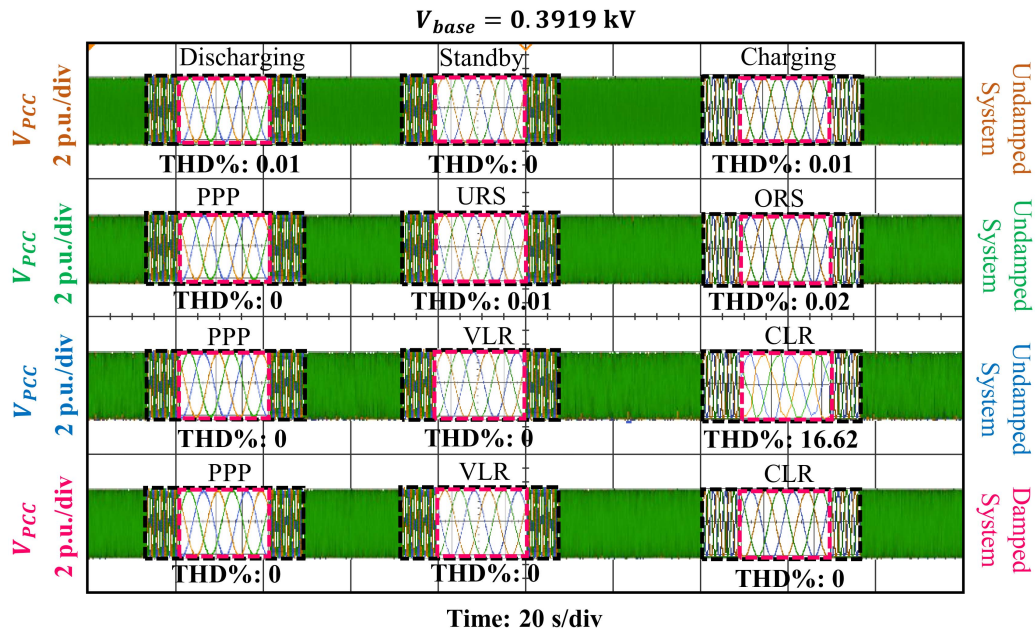


FIGURE 28. Performance of undamped and damped PCC voltage under various operating conditions.

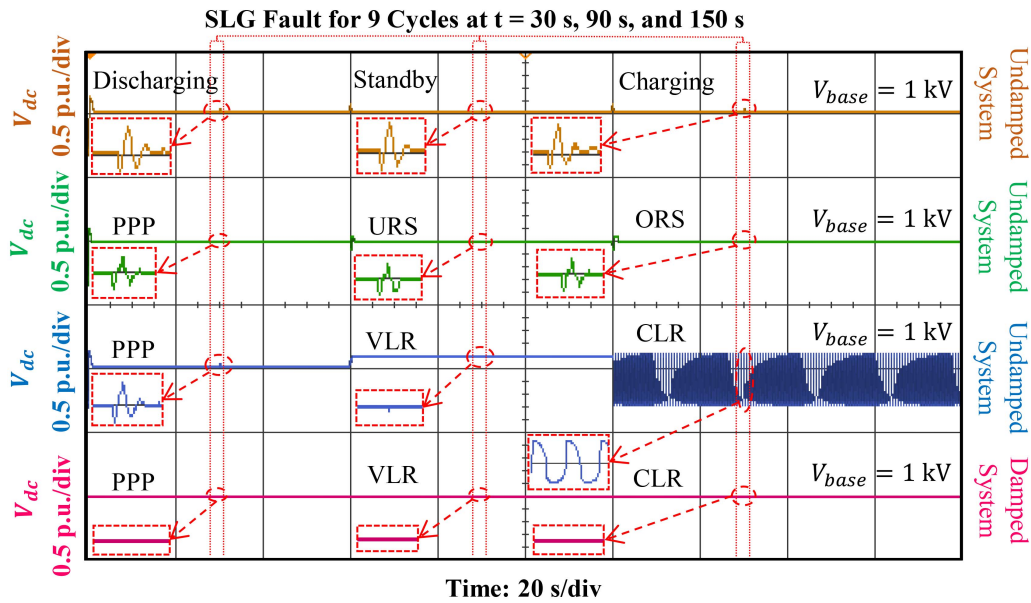


FIGURE 29. Performance of undamped and damped systems under an SLG fault at different operating conditions.

simulation offers several advantages [47], [48], [49], such as enabling system behavior testing under real-world conditions, achieving zero-overflow execution while considering potential time delays and external interactions, and expediting the prototyping process by providing immediate performance feedback. The OPAL-RT OP5600 platform is equipped with a Virtex-6 FPGA board connected to a four-channel 70 MHz oscilloscope and is seamlessly integrated with the MATLAB/Simulink environment through the SimPowerSystem toolbox. The PV generation source in this study consists of a

Canadian-solar array with 590 W modules, as specified in the datasheet [50]. The system parameters are provided in Table 2.

Recent advancements in real-time simulators are highlighted in [47], [48], and [49], focusing on the importance of model validation and the benefits of combining real-time and offline simulation modes. Real-time simulators generally fall into three categories:

1) Rapid-control-prototyping (RCP), where controllers for the plant are developed using a real-time simulator and connected to the physical plant;

2) Hardware-in-the-Loop (HIL) testing, which involves linking a physical controller to a virtual plant executed on a real-time simulator; and

3) SIL, which simulates both the controller and the plant in real-time within a unified and robust simulator environment. The RCP approach offers greater flexibility, faster implementation, and easier debugging than traditional controller prototyping techniques. In addition to RCP's benefits, HIL testing is cost-effective and enables early controller testing without the need for physical test benches. Conversely, SIL simulations outperform RCP and HIL simulations by preserving signal integrity without requiring inputs and outputs. In addition, timing synchronization with the external world is less critical since the controller and plant models operate within the same simulator. Therefore, SIL real-time simulations are used in this study.

A. PERFORMANCE OF THE UNDAMPED AND DAMPED SYSTEMS

In alignment with Figs. 8–10, Figs. 23–25 illustrate the undamped responses at $SCR = 2$ across various PV, WT, and BESS operating regions. The findings unequivocally affirm the system's stability and ability to meet the grid demand at diverse operating conditions of the battery, see Fig. 23, WT generator, see Fig. 24 and PV's VLR and PPP regions. However, Fig. 25 reveals that the system stability is violated within the PV generator's CLR region.

In contrast, the damping approach introduced in Fig. 26 demonstrates a notable enhancement in damping capability under the same operating conditions. Notably, this improvement is sustained even in the challenging CLR region, while stability is preserved in the VLR and PPP regions, with no adverse effects from the proposed damping approach.

Similar to the impedance-based stability analysis in Section IV-C, Fig. 27 illustrates the performance of the undamped and damped grid-tied hybrid systems under very weak grid conditions, i.e., $SCR=1.2$, and different operating regions. It is shown that the undamped system maintains stability across all battery and WT operating regions, as well as the PV's VLR and PPP regions. However, the undamped system exhibits LFO instability in the CLR region, which is mitigated by the proposed active damping technique.

B. PCC TOTAL HARMONIC DISTORTION ANALYSIS

Per the guidelines outlined in the IEEE Standard 519 [51], it is recommended that voltage distortions on the ac-side remain below 5% to ensure power quality. To assess the total harmonic distortion (THD) of the PCC voltage, a fast Fourier transform analysis is presented in Fig. 28, comparing scenarios with and without the proposed damping approach. The analysis distinctly shows a reduction in THD values to levels below 5%, signifying a significant improvement compared to the compromised quality observed in the absence of the damping approach.

C. PERFORMANCE UNDER FAULT CONDITIONS

Adhering to the grid codes for integrating renewable power systems [52], grid-tied renewable resources must exhibit resilience during voltage dips lasting nine cycles or less for a 60 Hz frequency, thereby achieving low-voltage-ride-through (LVRT). A single line-to-ground (SLG) fault is applied for nine cycles at the PCC in Fig. 1, at $t = 30, 90$, and 150 s. The fault timings are strategically positioned at the midpoint of each operating region, as shown in Fig. 29. The response of the dc-link voltage is meticulously examined during and after fault clearance, both in the presence and absence of the proposed active damping approach. While both damped and undamped systems override the faults across all operating regions, the damped system provides superior transient response and LFO damping.

VI. CONCLUSION

This article investigated the small-signal dynamics and stability of a weak grid-tied VSI-based hybrid energy system comprising a PV generator, a WT-PMSG, and a BESS. The study examined the influence of the dynamic resistances of these resources under weak grid conditions and across the BESS charging, discharging, and standby operational regions, as well as the PPP and off-PPP regions of the PV and WT. Small-signal analyses using the LSSM and ISAM models show that while the system maintains stability within the PV's VLR and PPP regions, as well as across all WT and BESS operating regions, instability emerges within the CLR region of the PV. In this region, the elevated dynamic resistance of the PV source interacts negatively with the high grid impedance under weak grid conditions, i.e., SCR between 1 and 2, resulting in LFO instability. A novel active damping approach is introduced to mitigate these adverse interaction dynamics and enhance overall system stability across all operating conditions. The proposed active damping method modifies the system's dc-link dynamics and incremental impedances to meet the Nyquist stability criterion. The key advantages of this approach include;

- 1) ease of design and implementation using conventional linear control techniques,
- 2) stable operation across various grid conditions without affecting steady-state performance,
- 3) improved LVRT capability, and
- 4) an efficient stabilization mechanism that eliminates the need for additional sensors, thereby reducing system complexity and cost.

Real-time and offline simulations validated the developed LSSM and ISAM models, interaction dynamics analysis, and the effectiveness of the proposed active damping method.

APPENDIX A

The equations of the control and ac and dc dynamics of the entire system in (1)–(23) are linearized as follows, with the

superscript “^o” represents a steady-state operating point

$$\begin{aligned}
 \begin{bmatrix} \Delta \dot{i}_d^g \\ \Delta \dot{i}_q^g \end{bmatrix} &= \overbrace{\begin{bmatrix} \frac{-(R_f+r_{on})}{L_f} & \omega^o \\ -\omega^o & \frac{-(R_f+r_{on})}{L_f} \end{bmatrix}}^{A_1} \begin{bmatrix} \Delta i_d^g \\ \Delta i_q^g \end{bmatrix} \\
 &+ \overbrace{\begin{bmatrix} \frac{1}{L_f} & 0 \\ 0 & \frac{1}{L_f} \end{bmatrix}}^{B_1} \begin{bmatrix} \Delta v_d^g \\ \Delta v_q^g \end{bmatrix} + \overbrace{\begin{bmatrix} \frac{-1}{L_f} & 0 \\ 0 & \frac{-1}{L_f} \end{bmatrix}}^{-B_1} \begin{bmatrix} \Delta v_d^g \\ \Delta v_q^g \end{bmatrix} \\
 &+ \overbrace{\begin{bmatrix} i_q^o \\ -i_d^o \end{bmatrix}}^{C_1} [\Delta \omega] \\
 \begin{bmatrix} \Delta \dot{i}_{gd}^g \\ \Delta \dot{i}_{gq}^g \end{bmatrix} &= \overbrace{\begin{bmatrix} \frac{-R_T}{L_T} & \omega^o \\ -\omega^o & \frac{-R_T}{L_T} \end{bmatrix}}^{A_2} \begin{bmatrix} \Delta i_{gd}^g \\ \Delta i_{gq}^g \end{bmatrix} + \frac{1}{N} \overbrace{\begin{bmatrix} \frac{1}{L_T} & 0 \\ 0 & \frac{1}{L_T} \end{bmatrix}}^{B_2} \begin{bmatrix} \Delta v_d^g \\ \Delta v_q^g \end{bmatrix} \\
 &+ \overbrace{\begin{bmatrix} \frac{-1}{L_T} & 0 \\ 0 & \frac{-1}{L_T} \end{bmatrix}}^{-B_2} \begin{bmatrix} \Delta e_{gd}^g \\ \Delta e_{gq}^g \end{bmatrix} + \overbrace{\begin{bmatrix} i_{gq}^o \\ -i_{gd}^o \end{bmatrix}}^{C_2} [\Delta \omega] \\
 \begin{bmatrix} \Delta v_d^g \\ \Delta v_q^g \end{bmatrix} &= \overbrace{\begin{bmatrix} 0 & \omega^o \\ -\omega^o & 0 \end{bmatrix}}^{A_3} \begin{bmatrix} \Delta v_d^g \\ \Delta v_q^g \end{bmatrix} + \overbrace{\begin{bmatrix} \frac{1}{C_f} & 0 \\ 0 & \frac{1}{C_f} \end{bmatrix}}^{B_3} \begin{bmatrix} \Delta i_d^g \\ \Delta i_q^g \end{bmatrix} \\
 &+ \frac{1}{N} \overbrace{\begin{bmatrix} \frac{-1}{C_f} & 0 \\ 0 & \frac{-1}{C_f} \end{bmatrix}}^{-B_3} \begin{bmatrix} \Delta i_{gd}^g \\ \Delta i_{gq}^g \end{bmatrix} + \overbrace{\begin{bmatrix} v_q^o \\ -v_d^o \end{bmatrix}}^{C_3} [\Delta \omega] \\
 \begin{bmatrix} \Delta v_{id}^c \\ \Delta v_{iq}^c \end{bmatrix} &= \overbrace{\begin{bmatrix} -g_p^i & -\omega^o L_f \\ \omega^o L_f & -g_p^i \end{bmatrix}}^{A_4} \begin{bmatrix} \Delta i_d^c \\ \Delta i_q^c \end{bmatrix} + \begin{bmatrix} \Delta \varphi^{G^i} \\ \Delta \varphi^{G^i} \end{bmatrix} + \begin{bmatrix} \Delta v_d^c \\ \Delta v_q^c \end{bmatrix} \\
 &+ \overbrace{\begin{bmatrix} g_p^i & 0 \\ 0 & g_p^i \end{bmatrix}}^{B_4} \begin{bmatrix} \Delta i_d^* \\ \Delta i_q^* \end{bmatrix} + \overbrace{\begin{bmatrix} -L_f i_q^o \\ L_f i_d^o \end{bmatrix}}^{C_4} [\Delta \omega] \\
 \begin{bmatrix} \Delta \varphi_d^{G^i} \\ \Delta \varphi_q^{G^i} \end{bmatrix} &= \overbrace{\begin{bmatrix} g_i^i & 0 \\ 0 & g_i^i \end{bmatrix}}^{A_5} \begin{bmatrix} \Delta i_d^* \\ \Delta i_q^* \end{bmatrix} + \overbrace{\begin{bmatrix} -g_i^i & 0 \\ 0 & -g_i^i \end{bmatrix}}^{-A_5} \begin{bmatrix} \Delta i_d^c \\ \Delta i_q^c \end{bmatrix} \\
 \begin{bmatrix} \Delta \delta^c \\ \Delta \varphi^{G^\delta} \end{bmatrix} &= \overbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}^{A_6} \begin{bmatrix} \Delta \delta^c \\ \Delta \varphi^{G^\delta} \end{bmatrix} + \overbrace{\begin{bmatrix} 0 & \frac{g_p^\delta}{v_d^o} \\ 0 & \frac{g_i^\delta}{v_d^o} \end{bmatrix}}^{B_6} \begin{bmatrix} \Delta v_d^c \\ \Delta v_q^c \end{bmatrix} \\
 [\Delta \omega] &= \overbrace{\begin{bmatrix} 0 & 1 \end{bmatrix}}^{C_6} \begin{bmatrix} \Delta \delta^c \\ \Delta \varphi^{G^\delta} \end{bmatrix} + \overbrace{\begin{bmatrix} 0 & \frac{g_p^\delta}{v_d^o} \\ 0 & \frac{g_i^\delta}{v_d^o} \end{bmatrix}}^{D_6} \begin{bmatrix} \Delta v_d^c \\ \Delta v_q^c \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \begin{bmatrix} \Delta v_d^c \\ \Delta v_q^c \end{bmatrix} &= \begin{bmatrix} \Delta v_d^g \\ \Delta v_q^g \end{bmatrix} + \overbrace{\begin{bmatrix} v_q^o & 0 \\ -v_d^o & 0 \end{bmatrix}}^{A_7} \begin{bmatrix} \Delta \delta^c \\ \Delta \varphi^{G^\delta} \end{bmatrix} \\
 \begin{bmatrix} \Delta i_d^c \\ \Delta i_q^c \end{bmatrix} &= \begin{bmatrix} \Delta i_d^g \\ \Delta i_q^g \end{bmatrix} + \overbrace{\begin{bmatrix} i_q^o & 0 \\ -i_d^o & 0 \end{bmatrix}}^{B_7} \begin{bmatrix} \Delta \delta^c \\ \Delta \varphi^{G^\delta} \end{bmatrix} \\
 \begin{bmatrix} \Delta v_{id}^g \\ \Delta v_{iq}^g \end{bmatrix} &= \begin{bmatrix} \Delta v_{id}^c \\ \Delta v_{iq}^c \end{bmatrix} + \overbrace{\begin{bmatrix} -v_{iq}^o & 0 \\ v_{id}^o & 0 \end{bmatrix}}^{C_7} \begin{bmatrix} \Delta \delta^c \\ \Delta \varphi^{G^\delta} \end{bmatrix} \\
 \begin{bmatrix} \Delta \varphi^{G^v} \\ \Delta I_{pv} \end{bmatrix} &= \overbrace{\begin{bmatrix} 0 & 0 \\ 0 & \frac{-R_{dc} - (P_{pv}^o / I_{pv}^{o^2})}{L_{dc}} \end{bmatrix}}^{A_8} \begin{bmatrix} \Delta \varphi^{G^v} \\ \Delta I_{pv} \end{bmatrix} \\
 &+ \overbrace{\begin{bmatrix} 0 & 0 \\ -\frac{1}{L_{dc}} & 0 \end{bmatrix}}^{B_8} \begin{bmatrix} \Delta V_{dc} \\ \Delta \varphi^{L^v} \end{bmatrix} + \overbrace{\begin{bmatrix} -g_i^v & 0 \\ 0 & 0 \end{bmatrix}}^{C_8} \begin{bmatrix} \Delta v_d^c \\ \Delta v_q^c \end{bmatrix} \\
 &+ \overbrace{\begin{bmatrix} g_i^v & 0 \\ 0 & 0 \end{bmatrix}}^{D_8} \begin{bmatrix} \Delta v_d^* \\ \Delta P_{pcc}^* \end{bmatrix} + \overbrace{\begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{L_{dc} I_{pv}^o} \end{bmatrix}}^{E_8} \begin{bmatrix} \Delta V_{dc}^* \\ \Delta P_{pv}^* \end{bmatrix} \\
 \begin{bmatrix} \Delta i_d^* \\ \Delta i_q^* \end{bmatrix} &= \overbrace{\begin{bmatrix} 0 & 0 \\ \frac{g_p^v}{1.5 v_d^o} & 0 \end{bmatrix}}^{A_9} \begin{bmatrix} \Delta v_d^c \\ \Delta v_q^c \end{bmatrix} + \overbrace{\begin{bmatrix} 0 & 0 \\ \frac{-1}{1.5 v_d^o} & 0 \end{bmatrix}}^{B_9} \begin{bmatrix} \Delta \varphi^{G^v} \\ \Delta I_{pv} \end{bmatrix} \\
 &+ \overbrace{\begin{bmatrix} 0 & \frac{1}{1.5 v_d^o} \\ \frac{-g_p^v}{1.5 v_d^o} & 0 \end{bmatrix}}^{C_9} \begin{bmatrix} \Delta v_d^* \\ \Delta P_{pcc}^* \end{bmatrix} \\
 \begin{bmatrix} \Delta \dot{i}_{sd} \\ \Delta \dot{i}_{sq} \end{bmatrix} &= \overbrace{\begin{bmatrix} \frac{-R_s}{L_{sd}} & \frac{P(\omega_s^{\text{magn}})^o L_{sq}}{L_{sd}} \\ \frac{-P(\omega_s^{\text{magn}})^o L_{sd}}{L_{sq}} & \frac{-R_s}{L_{sq}} \end{bmatrix}}^{A_{10}} \begin{bmatrix} \Delta i_{sd} \\ \Delta i_{sq} \end{bmatrix} \\
 &+ \overbrace{\begin{bmatrix} \frac{-1}{L_{sd}} & 0 \\ 0 & \frac{-1}{L_{sq}} \end{bmatrix}}^{B_{10}} \begin{bmatrix} \Delta v_{sd} \\ \Delta v_{sq} \end{bmatrix} \\
 &+ \overbrace{\begin{bmatrix} \frac{PL_{sq} i_{sq}^o}{L_{sd}} & 0 \\ \frac{-PL_{sd} i_{sd}^o + P \lambda_{\text{pm}}}{L_{sq}} & 0 \end{bmatrix}}^{C_{10}} \begin{bmatrix} \Delta \omega_s^{\text{magn}} \\ \Delta \varphi^{K^s} \end{bmatrix} \\
 \begin{bmatrix} \Delta v_{sd} \\ \Delta v_{sq} \end{bmatrix} &= \overbrace{\begin{bmatrix} k_p^i & P(\omega_s^{\text{magn}})^o L_{sq} \\ -P(\omega_s^{\text{magn}})^o L_{sd} & k_p^i \end{bmatrix}}^{A_{11}} \begin{bmatrix} \Delta i_{sd} \\ \Delta i_{sq} \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
& + \overbrace{\begin{bmatrix} -k_p^i & 0 \\ 0 & -k_p^i \end{bmatrix}}^{B_{11}} \begin{bmatrix} \Delta i_{sd}^* \\ \Delta i_{sq}^* \end{bmatrix} + \begin{bmatrix} \Delta \varphi_d^{K^i} \\ \Delta \varphi_q^{K^i} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} PL_{sq} i_{sq}^o & 0 \\ P\lambda_{pm} - PL_{sd} i_{sd}^o & 0 \end{bmatrix}}^{C_{11}} \begin{bmatrix} \Delta \omega_s^{\text{magn}} \\ \Delta \varphi^{K^s} \end{bmatrix} \\
\begin{bmatrix} \Delta \varphi_d^{K^i} \\ \Delta \varphi_q^{K^i} \end{bmatrix} & = \overbrace{\begin{bmatrix} k_i^i & 0 \\ 0 & k_i^i \end{bmatrix}}^{A_{12}} \begin{bmatrix} \Delta i_{sd} \\ \Delta i_{sq} \end{bmatrix} + \overbrace{\begin{bmatrix} -k_i^i & 0 \\ 0 & -k_i^i \end{bmatrix}}^{-A_{12}} \begin{bmatrix} \Delta i_{sd}^* \\ \Delta i_{sq}^* \end{bmatrix} \\
\begin{bmatrix} \Delta \omega_r^{\text{mech}} \\ \Delta \varphi^{K^s} \end{bmatrix} & = \overbrace{\begin{bmatrix} -B_r - [(T_b D_{sh})/\omega_b] & 0 \\ J_r & k_i^s \end{bmatrix}}^{A_{13}} \begin{bmatrix} \Delta \omega_r^{\text{mech}} \\ \Delta \varphi^{K^s} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} 0 & -1.5P\lambda_{pm} \\ 0 & 0 \end{bmatrix}}^{B_{13}} \begin{bmatrix} \Delta i_{sd} \\ \Delta i_{sq} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} T_b D_{sh} & T_b K_{sh} \\ J_r \omega_b & J_r \end{bmatrix}}^{C_{13}} \begin{bmatrix} \Delta \omega_{WT} \\ \Delta \theta_{sh} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} 0 & 0 \\ 0 & -k_i^s \end{bmatrix}}^{D_{13}} \begin{bmatrix} \Delta P_{WT} \\ \Delta \omega_r^{\text{mech}*} \end{bmatrix} \\
\begin{bmatrix} \Delta i_{sd}^* \\ \Delta i_{sq}^* \end{bmatrix} & = \overbrace{\begin{bmatrix} 0 & 0 \\ k_p^s - \eta[(T_b D_{sh})/\omega_b] & 1 \end{bmatrix}}^{A_{14}} \begin{bmatrix} \Delta \omega_r^{\text{mech}} \\ \Delta \varphi^{K^s} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} 0 & 0 \\ \eta[(T_b D_{sh})/\omega_b] & \eta T_b K_{sh} \end{bmatrix}}^{B_{14}} \begin{bmatrix} \Delta \omega_{WT} \\ \Delta \theta_{sh} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} 0 & 0 \\ 0 & -k_p^s \end{bmatrix}}^{C_{14}} \begin{bmatrix} \Delta P_{WT} \\ \Delta \omega_r^{\text{mech}*} \end{bmatrix} \\
\begin{bmatrix} \Delta \omega_{WT} \\ \Delta \theta_{sh} \end{bmatrix} & = \overbrace{\begin{bmatrix} -\omega_b P_{WT}^o & -D_{sh} & -\omega_b K_{sh} \\ 2H_{WT} T_b \omega_{WT}^o & 2H_{WT} & 2H_{WT} \\ 1 & 0 & 0 \end{bmatrix}}^{A_{15}} \begin{bmatrix} \Delta \omega_{WT} \\ \Delta \theta_{sh} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} D_{sh} & 0 \\ 2H_{WT} & 0 \end{bmatrix}}^{B_{15}} \begin{bmatrix} \Delta \omega_r^{\text{mech}} \\ \Delta \varphi^{K^s} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} \omega_b & 0 \\ 2H_{WT} T_b \omega_{WT}^o & 0 \end{bmatrix}}^{C_{15}} \begin{bmatrix} \Delta P_{WT} \\ \Delta \omega_r^{\text{mech}*} \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
\begin{bmatrix} \Delta \dot{I}_{bat} \\ \Delta \varphi^{L^i} \end{bmatrix} & = \overbrace{\begin{bmatrix} -R_{bat} - (P_{bat}^o/I_{bat}^o)^2 \\ L_{bat}^o & 0 \end{bmatrix}}^{A_{16}} \begin{bmatrix} \Delta I_{bat} \\ \Delta \varphi^{L^i} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} -\mathcal{D}^o & 0 \\ 0 & 0 \end{bmatrix}}^{B_{16}} \begin{bmatrix} \Delta V_{dc} \\ \Delta \varphi^{L^v} \end{bmatrix} + \overbrace{\begin{bmatrix} 0 \\ -l_i^i \end{bmatrix}}^{C_{16}} \begin{bmatrix} \Delta I_{bat}^* \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} -V_{dc}^o \\ L_{bat}^o & 0 \end{bmatrix}}^{D_{16}} [\Delta \mathcal{D}] + \overbrace{\begin{bmatrix} 1 \\ L_{bat}^o \end{bmatrix}}^{E_{16}} [\Delta P_{bat}] \\
[\Delta \mathcal{D}] & = \overbrace{\begin{bmatrix} l_p^i & 1 \end{bmatrix}}^{A_{17}} \begin{bmatrix} \Delta I_{bat} \\ \Delta \varphi^{L^i} \end{bmatrix} + \overbrace{\begin{bmatrix} -l_p^i \end{bmatrix}}^{B_{17}} [\Delta I_{bat}^*] \\
[\Delta I_{bat}^*] & = \overbrace{\begin{bmatrix} -l_p^v & 1 \end{bmatrix}}^{A_{18}} \begin{bmatrix} \Delta V_{dc} \\ \Delta \varphi^{L^v} \end{bmatrix} + \overbrace{\begin{bmatrix} l_p^v & 0 \end{bmatrix}}^{B_{18}} \begin{bmatrix} \Delta V_{dc}^* \\ \Delta P_{pv} \end{bmatrix} \\
\begin{bmatrix} \Delta \dot{V}_{dc} \\ \Delta \varphi^{L^v} \end{bmatrix} & = \overbrace{\begin{bmatrix} -1.5[(i_{sd}^o v_{sd}^o + i_{sq}^o v_{sq}^o) - (i_d^o v_{id}^o + i_q^o v_{iq}^o)] \\ C_{dc} V_{dc}^o{}^2 \\ -l_i^v \end{bmatrix}}^{A_{19}} \begin{bmatrix} \Delta V_{dc} \\ \Delta \varphi^{L^v} \end{bmatrix} \\
& \times \begin{bmatrix} \Delta V_{dc} \\ \Delta \varphi^{L^v} \end{bmatrix} + \overbrace{\begin{bmatrix} 0 & 0 \\ l_i^v & 0 \end{bmatrix}}^{B_{19}} \begin{bmatrix} \Delta V_{dc}^* \\ \Delta P_{pv} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}^{C_{19}} \begin{bmatrix} \Delta \varphi^{G^v} \\ \Delta I_{pv} \end{bmatrix} + \overbrace{\begin{bmatrix} 1.5v_{sd}^o & 1.5v_{sq}^o \\ C_{dc} V_{dc}^o & C_{dc} V_{dc}^o \end{bmatrix}}^{D_{19}} \begin{bmatrix} \Delta V_{dc}^* \\ \Delta P_{pv} \end{bmatrix} \\
& \times \begin{bmatrix} \Delta i_{sd} \\ \Delta i_{sq} \end{bmatrix} + \overbrace{\begin{bmatrix} 1.5i_{sd}^o & 1.5i_{sq}^o \\ C_{dc} V_{dc}^o & C_{dc} V_{dc}^o \end{bmatrix}}^{E_{19}} \begin{bmatrix} \Delta v_{sd} \\ \Delta v_{sq} \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} \mathcal{D}^o & 0 \\ 0 & 0 \end{bmatrix}}^{F_{19}} \begin{bmatrix} \Delta I_{bat} \\ \Delta \varphi^{L^i} \end{bmatrix} + \overbrace{\begin{bmatrix} l_{bat}^o \\ C_{dc} \end{bmatrix}}^{G_{19}} [\Delta \mathcal{D}] \\
& + \overbrace{\begin{bmatrix} -1.5v_{id}^o & -1.5v_{iq}^o \\ C_{dc} V_{dc}^o & C_{dc} V_{dc}^o \end{bmatrix}}^{H_{19}} \begin{bmatrix} \Delta i_{id}^g \\ \Delta i_{iq}^g \end{bmatrix} \\
& + \overbrace{\begin{bmatrix} -1.5i_{id}^o & -1.5i_{iq}^o \\ C_{dc} V_{dc}^o & C_{dc} V_{dc}^o \end{bmatrix}}^{I_{19}} \begin{bmatrix} \Delta v_{id}^g \\ \Delta v_{iq}^g \end{bmatrix}.
\end{aligned}$$

The states φ_{dq}^G , φ^{G^s} , φ^{G^v} , $\varphi_{dq}^{K^i}$, φ^{K^s} , φ^{L^i} , and φ^{L^v} represent the integral terms of the VSI ac current controller, the VSI

SRF-PLL controller, the VSI ac voltage controller, the VSR stator current controller, the VSR rotor speed controller, the 2-Q converter current controller, and the 2-Q converter dc-link voltage controller, respectively.

To verify the developed small-signal model in Section III-A, the following identity and zero matrices, along with the entries (block matrices) of the matrices A^{val} and B^{val} , are defined as follows:

$$I_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, Z_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, Z_{1 \times 2} = \begin{bmatrix} 0 & 0 \end{bmatrix},$$

$$Z_{1 \times 2}^T = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$a_{1,1} = A_1 + B_1 A_4$$

$$a_{1,3} = B_1 (B_4 A_9 + C_4 D_6) + C_1 D_6$$

$$a_{1,5} = B_1 (C_7 + A_4 B_7 + A_7 + B_4 A_9 A_7 + C_4 C_6) + C_1 C_6$$

$$a_{2,3} = B_2 / N + C_2 D_6$$

$$a_{3,3} = A_3 + C_3 D_6$$

$$a_{4,5} = -A_5 B_7 + A_5 A_9 A_7$$

$$a_{5,5} = A_6 + B_6 A_7$$

$$a_{7,7} = A_{10} + B_{10} A_{11}$$

$$a_{7,9} = C_{10} + B_{10} C_{11} + B_{10} B_{11} A_{14}$$

$$a_{11,11} = A_{16} + D_{16} A_{17}$$

$$a_{11,12} = B_{16} + C_{16} A_{18} + D_{16} B_{17} A_{18}$$

$$a_{12,1} = H_{19} + I_{19} A_4$$

$$a_{12,3} = I_{19} (I_{2 \times 2} + B_4 A_9 + C_4 D_6)$$

$$A^{\text{val}} = \begin{bmatrix} a_{1,1} & Z_{2 \times 2} & a_{1,3} & B_1 & a_{1,5} & B_1 B_4 B_9 & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} \\ Z_{2 \times 2} & A_2 & a_{2,3} & Z_{2 \times 2} & C_2 C_6 & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} \\ B_3 & -B_3 / N & a_{3,3} & Z_{2 \times 2} & C_3 C_6 & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} \\ -A_5 & Z_{2 \times 2} & A_5 A_9 & Z_{2 \times 2} & a_{4,5} & A_5 B_9 & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} \\ Z_{2 \times 2} & Z_{2 \times 2} & B_6 & Z_{2 \times 2} & a_{5,5} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} \\ Z_{2 \times 2} & Z_{2 \times 2} & C_8 & Z_{2 \times 2} & C_8 A_7 & A_8 & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & a_{7,7} & B_{10} & a_{7,9} & B_{10} B_{11} B_{14} & Z_{2 \times 2} & Z_{2 \times 2} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & A_{12} & Z_{2 \times 2} & -A_{12} A_{14} & -A_{12} B_{14} & Z_{2 \times 2} & Z_{2 \times 2} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & B_{13} & Z_{2 \times 2} & A_{13} & C_{13} & Z_{2 \times 2} & Z_{2 \times 2} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & B_{15} & A_{15} & Z_{2 \times 2} & Z_{2 \times 2} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & a_{11,11} & a_{11,12} \\ a_{12,1} & Z_{2 \times 2} & a_{12,3} & I_{19} & a_{12,5} & a_{12,6} & a_{12,7} & E_{19} & a_{12,9} & E_{19} B_{11} B_{14} & a_{12,11} & a_{12,12} \end{bmatrix}_{24 \times 24}$$

$$B^{\text{val}} = \begin{bmatrix} Z_{2 \times 2} & B_1 B_4 C_9 & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 1} \\ -B_2 & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 1} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 1} \\ Z_{2 \times 2} & A_5 C_9 & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 1} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 1} \\ Z_{2 \times 2} & D_8 & E_8 & Z_{2 \times 2} & Z_{2 \times 1} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & B_{10} B_{11} C_{14} & Z_{2 \times 1} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & -A_{12} C_{14} & Z_{2 \times 1} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & D_{13} & Z_{2 \times 1} \\ Z_{2 \times 2} & Z_{2 \times 2} & Z_{2 \times 2} & C_{15} & Z_{2 \times 1} \\ Z_{2 \times 2} & Z_{2 \times 2} & b_{11,3} & Z_{2 \times 2} & E_{16} \\ Z_{2 \times 2} & I_{19} B_4 C_9 & b_{12,3} & E_{19} B_{11} C_{14} & Z_{2 \times 1} \end{bmatrix}_{24 \times 9}$$

$$C^{\text{val}} = [\text{zeros}(1, 22) \quad 1 \quad 0]_{1 \times 24} \quad D^{\text{val}} = [\text{zeros}(1, 9)]_{1 \times 9}$$

$$\text{sys}^{\text{val}} = \text{ss}(A^{\text{val}}, B^{\text{val}}, C^{\text{val}}, D^{\text{val}}) = \text{sys}^{\text{uncomp}} = \text{ss}(A^{\text{uncomp}}, B^{\text{uncomp}}, C^{\text{uncomp}}, D^{\text{uncomp}})$$

$$\Delta x_1 = \begin{bmatrix} [\Delta i_d^g \quad \Delta i_q^g] & [\Delta i_{gd}^g \quad \Delta i_{gq}^g] & [\Delta v_d^g \quad \Delta v_q^g] & [\Delta \varphi_d^{G^i} \quad \Delta \varphi_q^{G^i}] & [\Delta \delta^c \quad \Delta \varphi^{G^s}] & [\Delta \varphi^{G^v} \quad \Delta I_{pv}] \end{bmatrix}_{1 \times 12}$$

$$\Delta x_2 = \begin{bmatrix} [\Delta i_{sd} \quad \Delta i_{sq}] & [\Delta \varphi_d^{K^i} \quad \Delta \varphi_q^{K^i}] & [\Delta \omega_r^{\text{mech}} \quad \Delta \varphi^{K^s}] & [\Delta \omega_{\text{WT}} \quad \Delta \theta_{\text{sh}}] & [\Delta I_{\text{bat}} \quad \Delta \varphi^{L^i}] & [\Delta V_{\text{dc}} \quad \Delta \varphi^{L^v}] \end{bmatrix}_{1 \times 12}$$

$$\Delta x = [\Delta x_1 \quad \Delta x_2]_{1 \times 24}^T$$

$$\Delta u = \begin{bmatrix} [\Delta e_{gd}^g \quad \Delta e_{gq}^g] & [\Delta v_d^* \quad \Delta P_{\text{pcc}}^*] & [\Delta V_{\text{dc}}^* \quad \Delta P_{\text{pv}}] & [\Delta P_{\text{WT}} \quad \Delta \omega_r^{\text{mech}*}] & \Delta P_{\text{bat}} \end{bmatrix}_{1 \times 9}^T$$

$$\Delta y = [\Delta V_{\text{dc}}] \text{ and } \text{TF} = \Delta V_{\text{dc}} / \Delta V_{\text{dc}}^* = \text{tf}(\text{sys}^{\text{uncomp}}(23, 5))$$

$$A^{\text{comp}} = \begin{bmatrix} \begin{bmatrix} A_{1,1}^{\text{uncomp}} & \dots & A_{1,11}^{\text{uncomp}} \\ \vdots & \ddots & \vdots \\ A_{10,1}^{\text{uncomp}} & \dots & A_{10,11}^{\text{uncomp}} \\ A_{11,1}^{\text{uncomp}} & \dots & A_{11,11}^{\text{uncomp}} \\ A_{12,1}^{\text{uncomp}} & \dots & A_{12,11}^{\text{uncomp}} \end{bmatrix} & \begin{bmatrix} A_{1,12}^{\text{uncomp}} \\ \vdots \\ A_{10,12}^{\text{uncomp}} \end{bmatrix} & [\text{zeros}(20, 1)] \\ \begin{bmatrix} B_{16} + (C_{16} + D_{16}B_{17})(A_{18} + B_{20}) \\ A_{19} + G_{19}B_{17}(A_{18} + B_{20}) \\ B_{20} \end{bmatrix} & \begin{bmatrix} (C_{16} + D_{16}B_{17})A_{20} \\ G_{19}B_{17}A_{20} \\ A_{20} \end{bmatrix} \end{bmatrix}_{25 \times 25}$$

$$B^{\text{comp}} = \begin{bmatrix} B^{\text{uncomp}} \\ \text{zeros}(1, 9) \end{bmatrix}_{25 \times 9} \quad C^{\text{comp}} = [\text{zeros}(1, 22) \quad 1 \quad \text{zeros}(1, 2)]_{1 \times 25} \quad D^{\text{comp}} = [\text{zeros}(1, 9)]_{1 \times 9}$$

$$\text{sys}^{\text{comp}} = \text{ss}(A^{\text{comp}}, B^{\text{comp}}, C^{\text{comp}}, D^{\text{comp}})$$

$$\begin{aligned} \Delta x_1 &= \begin{bmatrix} [\Delta i_d^g \quad \Delta i_q^g] & [\Delta i_{gd}^g \quad \Delta i_{gq}^g] & [\Delta v_d^g \quad \Delta v_q^g] & [\Delta \varphi_d^{G^i} \quad \Delta \varphi_q^{G^i}] & [\Delta \delta^c \quad \Delta \varphi^{G^\delta}] & [\Delta \varphi^{G^v} \quad \Delta I_{pv}] \end{bmatrix}_{1 \times 12} \\ \Delta x_2 &= \begin{bmatrix} [\Delta i_{sd} \quad \Delta i_{sq}] & [\Delta \varphi_d^{K^i} \quad \Delta \varphi_q^{K^i}] & [\Delta \omega_r^{\text{mech}} \quad \Delta \varphi^{K^s}] & [\Delta \omega_{WT} \quad \Delta \theta_{sh}] & [\Delta I_{bat} \quad \Delta \varphi^{L^i}] & [\Delta V_{dc} \quad \Delta \varphi^{L^v}] \end{bmatrix}_{1 \times 12} \\ \Delta x &= [\Delta x_1 \quad \Delta x_2 \quad \Delta X]_{1 \times 25}^T \\ \Delta u &= \begin{bmatrix} [\Delta e_{gd}^g \quad \Delta e_{gq}^g] & [\Delta v_d^* \quad \Delta P_{pcc}^*] & [\Delta V_{dc}^* \quad \Delta r_d] & [\Delta P_{WT} \quad \Delta \omega_r^{\text{mech}*}] & \Delta P_{bat} \end{bmatrix}_{1 \times 9}^T \\ \Delta y &= [\Delta V_{dc}] \text{ and } \text{TF} = \Delta V_{dc} / \Delta V_{dc}^* = t f(\text{sys}^{\text{comp}}(23, 5)). \end{aligned}$$

$$a_{12,5} = I_{19}(C_7 + A_4B_7 + A_7 + B_4A_9A_7 + C_4C_6)$$

$$a_{12,6} = C_{19} + I_{19}B_4B_9$$

$$a_{12,7} = D_{19} + E_{19}A_{11}$$

$$a_{12,9} = E_{19}C_{11} + E_{19}B_{11}A_{14}$$

$$a_{12,11} = F_{19} + G_{19}A_{17}$$

$$a_{12,12} = A_{19} + G_{19}B_{17}A_{18}$$

$$b_{11,3} = C_{16}B_{18} + D_{16}B_{17}B_{18}$$

$$b_{12,3} = B_{19} + G_{19}B_{17}B_{18}.$$

$$+ \begin{bmatrix} 0 & 0 \\ 0 & -\frac{I_{pv}^o}{L_{dc}} \end{bmatrix} \begin{bmatrix} \Delta V_{dc}^* \\ \Delta r_d \end{bmatrix}$$

APPENDIX B

The state space model of the uncompensated hybrid PV-WT-BESS system is similar to that of the verification model, i.e., $\text{sys}^{\text{uncomp}} = \text{sys}^{\text{val}}$, but with a different mathematical representation for V_{pv} in (11), i.e., $V_{pv} = -r_d I_{pv}$ instead of $V_{pv} = P_{pv}/I_{pv}$, to study the impact of the dynamic resistance of the PV generator. In the validation input vector, i.e., Δu , the input ΔP_{pv} is replaced with Δr_d .

The modified equation is linearized as follows:

$$\begin{aligned} \begin{bmatrix} \Delta \varphi^{G^v} \\ \Delta I_{pv} \end{bmatrix} &= \begin{bmatrix} 0 & 0 \\ 0 & -\frac{r_d^o - R_{dc}}{L_{dc}} \end{bmatrix} \begin{bmatrix} \Delta \varphi^{G^v} \\ \Delta I_{pv} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -\frac{1}{L_{dc}} & 0 \end{bmatrix} \begin{bmatrix} \Delta V_{dc} \\ \Delta \varphi^{L^v} \end{bmatrix} \\ &+ \begin{bmatrix} -g_i^v & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta v_d^c \\ \Delta v_q^c \end{bmatrix} + \begin{bmatrix} g_i^v & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta v_d^* \\ \Delta P_{pcc}^* \end{bmatrix} \end{aligned}$$

APPENDIX C

An additional state, i.e., X , is introduced to represent the proposed damping method in (24)–(25). The state space model of the compensated hybrid PV-WT-BESS system is similar to the uncompensated state space model, $\text{sys}^{\text{comp}} = \text{sys}^{\text{uncomp}}$, with modifications to (20).

The compensated model includes the linearized equations of (24)–(25)

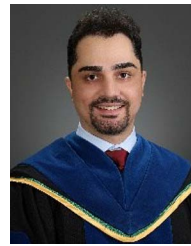
$$\begin{aligned} [\Delta I_{bat}^*] &= \begin{bmatrix} A_{18} \\ -I_p^v & 1 \end{bmatrix} \begin{bmatrix} \Delta V_{dc} \\ \Delta \varphi^{L^v} \end{bmatrix} + \begin{bmatrix} B_{18} \\ I_p^v & 0 \end{bmatrix} \begin{bmatrix} \Delta V_{dc}^* \\ \Delta P_{pv}^* \end{bmatrix} + [\Delta \dot{X}] \\ [\Delta \dot{X}] &= \begin{bmatrix} A_{20} \\ -1 & \tau \end{bmatrix} [\Delta X] + \begin{bmatrix} B_{20} \\ \alpha & 0 \end{bmatrix} \begin{bmatrix} \Delta V_{dc} \\ \Delta \varphi^{L^v} \end{bmatrix}. \end{aligned}$$

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